Density-Based Dynamic Ensemble Learning Algorithm via Gaussian Mixture Model Clustering for Enhanced Predictive Analytics

Abstract—Dynamic ensemble learning enhances predictive accuracy by combining multiple classifiers, but existing methods struggle to balance diversity with generalization and to adapt when data distributions shift. This paper introduces DDEL-GMM, a Density-Based Dynamic Ensemble Learning Algorithm that uses Gaussian Mixture Models to partition the training space into overlapping subsets reflecting local data structure, trains distinct base classifiers on each subset, and selects classifiers per test instance via density-weighted competence estimation. Principal Component Analysis (157 components retaining 95% of variance) keeps clustering tractable in high dimensions. Evaluated on the UCI Human Activity Recognition dataset with logistic regression, SVM, KNN, and decision tree base learners, DDEL-GMM significantly outperforms published dynamic ensemble selection methods including DELAK ($p = 0.002$), META-DES ($p = 0.019$), KNORA-U, and KNORA-E ($p \leq 0.002$), while matching the strongest linear and bagged baselines on stationary data. Under simulated concept drift, DDEL-GMM degrades 3.6× less than DELAK and 8.7× less than a single logistic regression, demonstrating that the density-based weighting confers robustness to distributional change. Ablation studies isolate the contributions of GMM clustering (+1.1 F-score points over K-means) and density-based weighting, and show that the framework reduces fold-to-fold dispersion by an order of magnitude relative to distance-based selection within the same clustering framework.

Index Terms—Dynamic Ensemble Learning, Gaussian Mixture Models, Density-Based Ensemble Learning, Predictive Analytics, Model Personalization, Decision-Making Efficiency.

I. Introduction

THE advent of dynamic ensemble classifiers heralds a new epoch in machine learning, addressing critical needs across sectors such as medical diagnostics [1], cybersecurity [2], and more. Recent advancements have shown that these classifiers significantly enhance the accuracy and robustness of human activity recognition (HAR) systems, as well as medical diagnostics, particularly in fields like cardiovascular disease detection using novel ensemble classifiers [3]. For instance, evolutionary dual-ensemble class imbalance learning methods have demonstrated superior performance in recognizing diverse human actions by optimizing the combination of base classifiers [4]. Ensemble learning, by synergizing multiple classifiers, offers unparalleled robustness and accuracy, far surpassing the capabilities of individual classifiers [5]–[9]. This approach, known as Multiple Classifier Systems (MCS), leverages diverse models for enhanced prediction reliability. Specifically, Dynamic Ensemble Selection (DES) refines this concept by customizing classifier combinations for each test instance, thus ensuring adaptability and precision in predictive analytics [10], [11]. The sophistication of ensemble methods is evident in their application to complex tasks, where they have consistently outperformed single-model approaches [12], [13]. Despite the advancements, ensemble learning faces significant challenges. Key among these is the need for personalized predictions for scenarios such as fluctuating market trends or evolving cybersecurity threats [14]. Traditional ensemble methods struggle with the nuanced balance between classifier diversity and ensemble generalization, a barrier to harnessing their full potential [15]. Furthermore, the dilemma of maintaining robustness without increasing ensemble complexity presents a persistent challenge in dynamic selection methods. These issues underscore the necessity for innovative solutions that can adapt to variable data conditions while maintaining accuracy and efficiency [14].

Before presenting our contribution, we briefly recap how DES methods address these challenges and where they remain limited. DES tailors the classifier combination to each test instance by estimating classifier competence within a local region of the feature space, typically the neighbourhood of the test point identified through a k-nearest-neighbour rule [16], [17]. Because a fixed distance-based neighbourhood is sensitive to the local structure of the data, subsequent work has refined how the region of competence is defined and how competence is scored: through meta-learning over classifier behaviour [18], prediction-confidence thresholds [10], distance-based reweighting [11], and distribution-aware selection [13]. This line of refinement continues in recent work: graph neural networks have been used to model competence over high-dimensional overlapped data [19], self-generating prototypes combined with a meta-classifier have been proposed to stabilise the competence estimate [20], and instance-hardness measures have been coupled with the region-of-competence strategy under Bayesian optimisation [21]. Despite these advances, competence estimation in these methods remains anchored to a locally defined neighbourhood or an additional meta-learning layer, which leaves two challenges open: the geometric definition of the competence region is sensitive to sparse or overlapping data, and the diversity–generalization trade-off [15] together with the robustness-versus-complexity tension [14] persists. These limitations motivate estimating competence directly from a generative model of the training-data distribution rather than from a fixed local neighbourhood.

Our proposed model, the Density-Based Dynamic Ensemble Learning algorithm via Gaussian Mixture Model clustering (DDEL-GMM), addresses these limitations by replacing neighbourhood-based competence estimation with a probabilistic, density-based one grounded in a Gaussian Mixture Model of the training data. The main contributions of this work are as follows:

1) GMM-based partitioning of the training data. We use a Gaussian Mixture Model to softly partition the training set into K subsets, each capturing a component of the underlying data distribution, and train a base learner specialised to each subset. Unlike hard, distance-based partitions, this produces a set of inherently diverse and locally specialised classifiers from the outset.
2) Density-based dynamic weighting at inference. For each test instance, the contribution of every base learner is weighted by the Gaussian density of the instance under the corresponding GMM component, so competence is assessed from the estimated data distribution rather than from a fixed k-nearest-neighbour region.
3) Empirical evaluation across base learners and aggregation schemes. We instantiate the framework with logistic regression, support vector machine, k-nearest-neighbour, and decision-tree base learners, and evaluate the density-based weighting against the DistE, AvgE, and MaxE aggregation baselines, analysing the effect of the number of clusters and the density weighting parameter.

This study leverages a complex dataset with high-dimensional characteristics, generated from wearable sensor technology, to rigorously evaluate the performance of the proposed dynamic ensemble learning model. The dataset originates from recordings of human activity, captured using accelerometers and gyroscopes integrated into smartphones. These sensors meticulously track the subtle movements and changes in orientation as subjects engage in various physical activities, providing a rich source of information for activity recognition tasks.

Each recording in the dataset corresponds to a distinct activity performed by a subject and encapsulates a time series of sensor readings. At each time step within a recording, the accelerometers and gyroscopes measure linear acceleration and angular velocity, respectively, across three orthogonal axes (X, Y, and Z). This intricate sampling process generates a high-dimensional feature space, where each feature represents the magnitude of a specific sensor measurement (acceleration or angular velocity) along a particular axis at a given time instance. To address the computational challenges of this high-dimensional space and to ensure model stability, Principal Component Analysis (PCA) was applied as a preprocessing step. The analysis showed that retaining 157 principal components was sufficient to preserve 95% of the variance, enabling efficient and effective clustering and classification within the DDEL-GMM framework.

While Gaussian Mixture Models offer powerful probabilistic modeling capabilities, it is well-established that GMM parameter estimation via Expectation-Maximization becomes computationally challenging and numerically unstable in high-dimensional settings [22]. Specifically, the curse of dimensionality affects the estimation of the full covariance matrix of each component: with p features and K components, the number of parameters grows as $\mathcal{O}(Kp^2)$, and sample complexity requirements scale accordingly. This places GMM-based clustering beyond practical reach when p exceeds several hundred without intervention. However, our design directly addresses this limitation through dimensionality reduction as a preprocessing step. By reducing the feature space from the original high-dimensional sensor data to 157 principal components (retaining 95% of variance), we map the problem into a regime where covariance estimation is feasible. The reduction is substantial: at the ambient dimensionality of $p = 561$, a $K = 5$ full-covariance mixture would require 788,205 covariance parameters, whereas at $p = 157$ it requires 62,015. We do not claim that $p = 157$ is comfortably low-dimensional for full-covariance EM—with roughly 9,269 training samples per fold the sample-to-parameter ratio is approximately 0.15, so the covariance estimates remain regularised rather than well-determined, and Section IV-F quantifies empirically how the clustering advantage attenuates as this ratio deteriorates. Two properties of our design keep the method stable in this regime. First, the density-based weighting depends on the relative ordering of component densities across clusters rather than on absolute likelihood values, so it tolerates imprecise covariance estimates that would compromise likelihood-based model selection. Second, regularised covariance estimation bounds the smallest eigenvalue away from zero, preventing the singular solutions that make unregularised high-dimensional EM diverge. This preprocessing strategy is well-supported in the literature: recent surveys of dimensionality reduction techniques highlight PCA as a fundamental and effective enabler of clustering and mixture-model learning in high-dimensional domains [23]. PCA prior to GMM clustering thus converts an infeasible estimation problem into a tractable, regularised one, at the cost of a covariance estimate whose precision we characterise rather than assume.

The raw sensor data, while rich in information, is inherently prone to noise and uncertainty. This stems from a confluence of factors inherent to the nature of wearable sensing:

- Sensor Imperfections: Like all physical instruments, accelerometers and gyroscopes exhibit limitations in their sensitivity and accuracy. Minute variations in sensor readings can arise from internal electronic noise, temperature fluctuations, and manufacturing inconsistencies.
- Environmental Interference: The surrounding environment can introduce spurious signals that contaminate the sensor readings. Electromagnetic interference from nearby electronic devices, vibrations from the

floor or other surfaces, and even changes in air pressure can all contribute to noise in the data.
- Placement and Movement Variability: The placement of the smartphone on the subject's body (e.g., pocket, belt, hand) and variations in the manner of movement can significantly influence the sensor readings. Slight shifts in the device's position or changes in the execution of an activity introduce inconsistencies into the collected data.

These inherent challenges, particularly the variability arising from sensor placement and individual movement styles, underscore the limitations of a rigid, single model classification model. This is where the proposed dynamic ensemble model demonstrates its strength. By employing Gaussian Mixture Models (GMMs), the model effectively captures the underlying data distribution, accounting for its multi-modality and effectively distinguishing meaningful patterns from random fluctuations. Moreover, the model's density-based selection strategy ensures that for each test instance, the most appropriate base learners, trained on relevant data partitions, are selected for prediction. This dynamic and context-aware approach enhances the model's robustness and accuracy, enabling it to achieve high performance even when faced with the complexities of real-world sensor data.

This paper is organized as follows: Section II provides a comprehensive review of existing literature in ensemble learning, dynamic ensemble selection, and related clustering approaches, positioning our work within this context, highlighting the challenges addressed, and underscoring its contributions. Section III details the proposed DDEL-GMM framework, elaborating on the GMM-based data clustering and the density-based dynamic selection mechanism. Section IV presents the experimental setup, discusses the evaluation results comparing DDEL-GMM with other methods, and analyzes the impact of key parameters. Finally, Section V concludes the paper, summarizing the findings and suggesting potential directions for future research.

II. Literature Review

While ensemble learning and dynamic ensemble selection (DES) have shown considerable promise [10], [11], [17], significant challenges remain, particularly in effectively handling complex data distributions, adapting predictions to individual instances, and efficiently balancing classifier diversity with generalization [14], [15]. This review examines existing approaches and positions our proposed DDEL-GMM, which addresses these challenges through a novel density-based framework leveraging Gaussian Mixture Models, by highlighting its differences from state-of-the-art methods and underscoring its contributions.

A. Ensemble Learning and Dynamic Ensemble Selection

Ensemble learning marks a significant stride in machine learning, particularly in the context of predictive modeling. Combining multiple models, such as through Bagging, Random Subspace Method, and Random Forest, ensemble learning approaches have improved predictive accuracy and robustness [17]. The diversification among base models is key to the success of these methods. Furthermore, dynamic ensemble selection techniques have evolved, allowing for the adaptation of model selection based on each test sample's specific characteristics, thereby offering more personalized predictions [16]. This is particularly useful in applications like breast cancer detection, where an ensemble of deep learning networks has been shown to improve diagnostic accuracy using consensus-adaptive weighting techniques [24].

The DeepCluE approach employs a convolutional neural network to generate diverse clusterings from multiple layers, effectively selecting the most informative ones for final clustering [25]. Similarly, the probabilistic combination of non-linear eigenprojections for ensemble classification enhances the reliability of classifications by integrating a probabilistic support vector machine (SVM) approach, allowing for better handling of uncertainty in predictions [26].

These examples illustrate the ongoing efforts to enhance ensemble methods, yet many still grapple with optimal diversity generation and adaptive selection in highly heterogeneous data. Our DDEL-GMM contributes by proposing a GMM-based data partitioning that aims to create inherently diverse and specialized base classifiers, followed by a density-based selection mechanism that adapts to local data characteristics more fluidly than some consensus or static weighting schemes.

B. Ensemble Pruning and Clustering-based Approaches

Ensemble pruning optimizes the selection of classifiers within an ensemble to balance efficiency and performance. Diverse strategies, including ranking-based methods and optimization-based methods like Optimum-Path Forest (OPF) ensemble pruning [27], illustrate the evolving landscape of ensemble pruning. Additionally, clustering-based approaches, which organize similar classifiers and select the most representative ones for the ensemble, add another dimension to enhancing the efficiency of ensemble learning systems [17].

While pruning focuses on selecting from an existing pool, and traditional clustering often groups classifiers post-training, DDEL-GMM integrates clustering directly into the training data partitioning phase using GMMs. This proactive approach aims to build a diverse and competent ensemble from the outset, rather than pruning less effective members later, addressing the challenge of efficiently constructing high-performing ensembles. The contribution here is a more integrated clustering-for-diversity strategy rather than clustering-for-pruning.

C. Dynamic Classifier Selection Techniques

Dynamic classifier selection techniques, focusing on estimating and utilizing the competence levels of classifiers, have seen significant advancements. Approaches like META-DES and Dynamic Selection on Complexity (DSOC) have emphasized the importance of the region of competence in classifier selection, leading to more accurate and tailored predictions. The refinement of these techniques, considering both correctly-classified and misclassified samples, has contributed to improved accuracy in dynamic selection systems, making them more effective in diverse machine learning applications [17].

Many DES techniques define the region of competence based on K-Nearest Neighbors or local instance similarity. DDEL-GMM defines competence based on the probability density of a test instance with respect to GMM-defined clusters. This is a key difference, as it moves from instance-based distance to a probabilistic, model-based density estimation for competence assessment, potentially offering more robust selections in sparse or noisy regions. The contribution is a density-informed competence estimation for DES.

D. Recent Advancements and Current Limitations

Guo et al. [28] exposed the shortcomings of traditional scoring systems like SAPS II and SOFA, and single machine learning models in critical care settings [29], [30]. This work introduced the Dynamic Ensemble Learning Algorithm based on K-means (DELAK) method, employing K-means sampling and dynamic ensemble selection to address these issues, an approach in line with current trends in ensemble methods for enhanced predictive performance [31], [32]. However, DELAK, like many other distance-based dynamic ensemble methods, struggled to capture the intricacies of complex data distributions and personalize predictions for new test instances, a limitation echoed in contemporary research [33], [34].

DELAK's reliance on K-means, which typically forms spherical clusters and can be sensitive to initialization, is a key point of divergence for DDEL-GMM. Our approach uses GMMs, which can model non-spherical clusters and provide a probabilistic assignment, addressing the challenge of representing complex data distributions more accurately. The contribution of DDEL-GMM here is the use of a more flexible clustering algorithm (GMM) to better partition complex data spaces for training specialized classifiers, leading to improved personalization. This choice is grounded in the maturity of Gaussian mixture learning, including efficient hierarchical-clustering-initialized EM estimation [35] and recent theoretical guarantees for robustly learning general mixtures of Gaussians [36].

Further contributions, such as the META-DES.Oracle framework in [18], extended the scope by utilizing meta-learning for classifier selection. While groundbreaking, this approach grappled with potential overfitting and computational complexities stemming from its intricate meta-feature framework and the optimization process using PSO-BP optimization algorithm, which integrates Particle Swarm Optimization (PSO) with the Backpropagation (BP) neural network training method [37]. Additionally, its reliance on linear classifiers limited its effectiveness in scenarios with non-linear relationships. DDEL-GMM, in contrast, does not employ a meta-learning layer for selection but rather a direct density-based assessment, potentially reducing complexity and overhead compared to intricate meta-feature frameworks. It also inherently supports non-linear base classifiers without modification to the selection mechanism itself, addressing the challenge of applicability to diverse base models. Recent ensemble learning advancements have focused on overcoming these limitations by exploring diverse methods like bagging, boosting, and decision tree ensembles, which have been shown to offer more robust solutions in handling complex data patterns [30]. These techniques, along with novel approaches such as ensemble methods for dirty data, provide a framework for integrating multiple learning algorithms and addressing the challenges observed in META-DES.Oracle, particularly ensuring model generalization [38].

Similarly, Wang et al. [39] and Zhang et al. [11] introduced novel selection methods, yet faced challenges in the sensitivity of unsupervised ensemble selection to the influence of class labels and the limitations of distance metrics for assessing classifier competence. These issues are also reflected in studies focusing on unsupervised data selection and ensemble classifiers, which further emphasize the complexities in selecting and combining multiple learning models without the guidance of labeled data [40]. To address such challenges related to unsupervised selection and reliance on distance metrics, DDEL-GMM employs a distinct strategy. It leverages GMM-identified data structures to inform the creation of specialized base learners from labeled data and subsequently uses a density-based mechanism for dynamic classifier selection. This approach aims to provide a more robust competence assessment by moving beyond simple distance metrics and more effectively incorporating insights from the underlying data distribution, even when initial structures are found unsupervisedly, thereby offering an alternative to the sensitivities noted in prior works. Recent advancements in unsupervised ensemble classification, particularly in handling data dependencies and using clustering ensembles for feature selection, provide new perspectives and potential solutions to these challenges [41], [42], [43].

Sowkarthika et al. [44] tackled the issue of class overlapping and noise, highlighting the inadequacy of existing ensemble methods in handling these complexities. Additionally, Perez-Gallego et al. [45] addressed the unique challenges in quantification tasks, proposing a dynamic ensemble method integrating data complexity measures. However, adaptability to data variability data environments remained a challenge for these approaches, an issue further explored in recent works such as [46] dynamic neural network ensemble algorithm, which effectively balances diversity and accuracy across various inputs. New insights into dynamic ensemble methods for data streams are provided by [47], who discuss the integration of drift detection algorithms and dynamic updates, crucial for real-time data processing. The GMM component of DDEL-GMM is inherently suited to model overlapping distributions through probabilistic assignments, and the density-based selection is designed to be adaptive. This addresses the challenge of handling class overlap more effectively than methods relying on hard partitioning, as the probabilistic nature of GMMs can model these overlapping regions. The contribution is enhanced robustness in scenarios with significant class overlap.

In summary, while existing dynamic ensemble methods like K-means-based (e.g., DELAK) and meta-learning based (e.g., META-DES.Oracle) have advanced the field, they often face limitations in handling highly complex and overlapping data distributions, potential computational overhead, or sensitivity to the geometric definition of competence regions. Our proposed DDEL-GMM differentiates itself by integrating Gaussian Mixture Models for sophisticated data clustering—capturing multi-modal and non-spherical data patterns—and a subsequent density-based dynamic selection strategy. This approach directly addresses the challenges of achieving robust personalization and adaptability in complex, noisy environments. The primary contribution of DDEL-GMM to the literature is thus a more flexible and statistically grounded framework for dynamic ensemble learning, designed to improve predictive accuracy and decision-making efficiency by more accurately modeling data sub-structures and employing a nuanced, density-informed mechanism for classifier selection, especially for data exhibiting intricate structures, as often encountered in real-world applications like sensor-based activity recognition.

III. Methodology

A. The proposed DDEL-GMM framework

This paper proposes a novel dynamic ensemble model, the Density-Based Dynamic Ensemble Learning algorithm via Gaussian Mixture Model clustering (DDEL-GMM). DDEL-GMM is designed to enhance predictive accuracy in providing personalized predictions. By employing Gaussian Mixture Models (GMM) for clustering and a density-based decision-making approach, the model aims to achieve a more granular analysis of complex data distributions and provide a robust framework for classifier competence estimation.

Figure 1 traces the method in three stages. The empirical behaviour it produces—which aggregation rule wins, why, and where the mechanism runs out—is reported in Section IV and summarised in Figure 7.

As illustrated in Figure 2, the training phase of DDEL-GMM involves leveraging GMM to effectively capture the underlying structure of the data. This approach facilitates the creation of specialized classifiers that are adept at recognizing specific data patterns within complex and overlapping distributions.

Furthermore, as depicted in Figure 3, the generalization phase employs a density-based dynamic ensemble selection mechanism. This ensures that, during prediction, the most relevant classifiers contribute more significantly based on the density estimation of the input data point, thereby enhancing the model's overall performance and robustness.

We introduce an advanced ensemble learning strategy that is deeply rooted in the principles of data density and specialization. Our proposed methodology consists of two interrelated components: the Gaussian Mixture Model (GMM) Sampling and the Density-based Dynamic Ensemble.

B. GMM Clustering & Sampling

Traditional ensemble techniques often create subsets of data either randomly or using simple heuristics. While effective for many datasets, the increasing complexity and multimodal nature of modern datasets necessitate a more sophisticated approach. Enter GMM Sampling, a method designed to capture the intricate structures within datasets and thereby enhance the versatility of ensemble models.

Motivation: The driving force behind employing GMM in the ensemble paradigm is its ability to recognize and capture underlying patterns within datasets. Given that many datasets exhibit a multimodal distribution, GMM's capability to model data as a mixture of several Gaussian distributions becomes invaluable. By ensuring that each base classifier in the ensemble is trained on data that mirrors these Gaussian components, we bolster the ensemble's capacity to recognize and respond to subtle data structures.

GMM Overview: Mathematically, a Gaussian Mixture Model represents data as a weighted combination of M Gaussian distributions as shown in Equation 1.

$$p(\mathbf{x}) = \sum_{i=1}^{M} \pi_i \mathcal{N}\left(\mathbf{x} \mid \mu_i, \Sigma_i\right) \tag{1}$$

Where:

- x is a data point.
- π_i denotes the mixture weight of the i^{th} Gaussian.
- $\mathcal{N}\left(\mathbf{x} \mid \mu_i, \Sigma_i\right)$ is the Gaussian distribution with parameters mean μ_i and covariance Σ_i.

The Gaussian distribution $\mathcal{N}\left(\mathbf{x} \mid \mu_i, \Sigma_i\right)$ is defined as shown in Equation 2:

$$\begin{aligned}\mathcal{N}\left(\mathbf{x} \mid \mu_i, \Sigma_i\right) = & \frac{1}{(2\pi)^{d/2}\left|\Sigma_i\right|^{1/2}} \\ & \times \exp\left(-\frac{1}{2}\left(\mathbf{x}-\mu_i\right)^T \Sigma_i^{-1}\left(\mathbf{x}-\mu_i\right)\right)\end{aligned} \tag{2}$$

where d is the dimensionality of x.

Parameter Estimation via Expectation-Maximization (EM): The mixture parameters $\{\pi_i, \mu_i, \Sigma_i\}_{i=1}^{K}$ are estimated with the standard Expectation–Maximization (EM) algorithm [48], which iteratively maximizes the data log-likelihood by alternating an E-step and an M-step until the parameters converge. The E-step computes

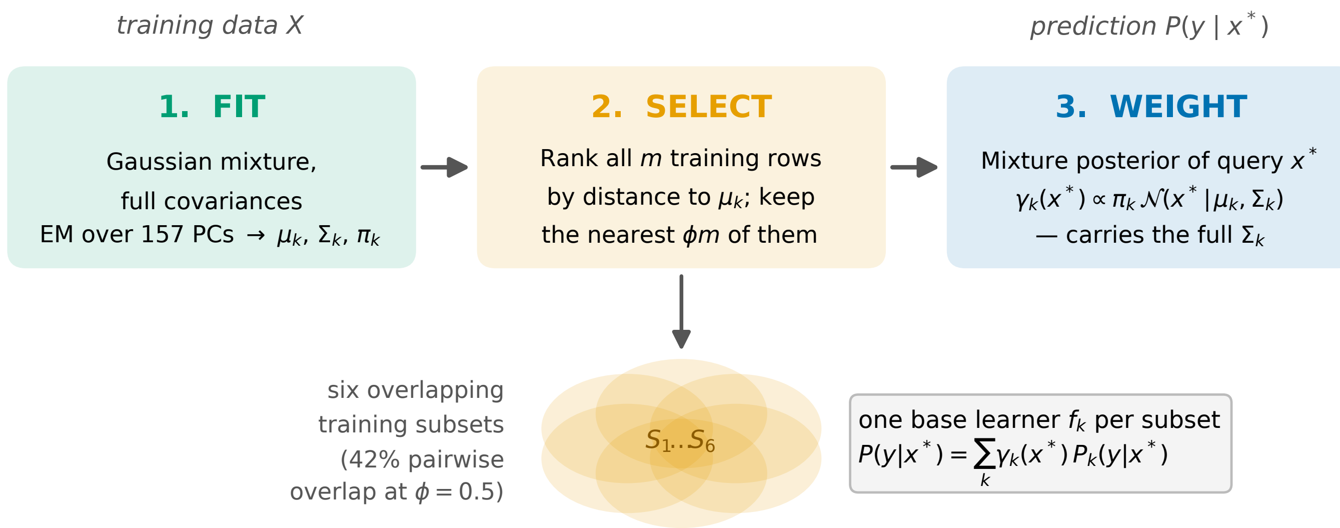

Fig. 1. The DDEL-GMM pipeline. Fit: a Gaussian mixture with full covariance matrices is estimated by EM over the 157 retained principal components, yielding μ_k, Σ_k and π_k for $K = 6$ components. Select: the training rows are ranked by distance to each μ_k and the nearest ϕm are retained, where ϕ is a fraction of the entire training set rather than of a single cluster; the six resulting subsets therefore overlap (42% pairwise at the cross-validated operating point $\phi = 0.5$). Weight: at inference each base learner f_k is weighted by the mixture posterior $\gamma_k(x^*) \propto \pi_k \mathcal{N}(x^* \mid \mu_k, \Sigma_k)$, so the weighting—unlike inverse-distance schemes—is anisotropic and carries the full component covariance. Dataset: UCI HAR, 10,299 instances.

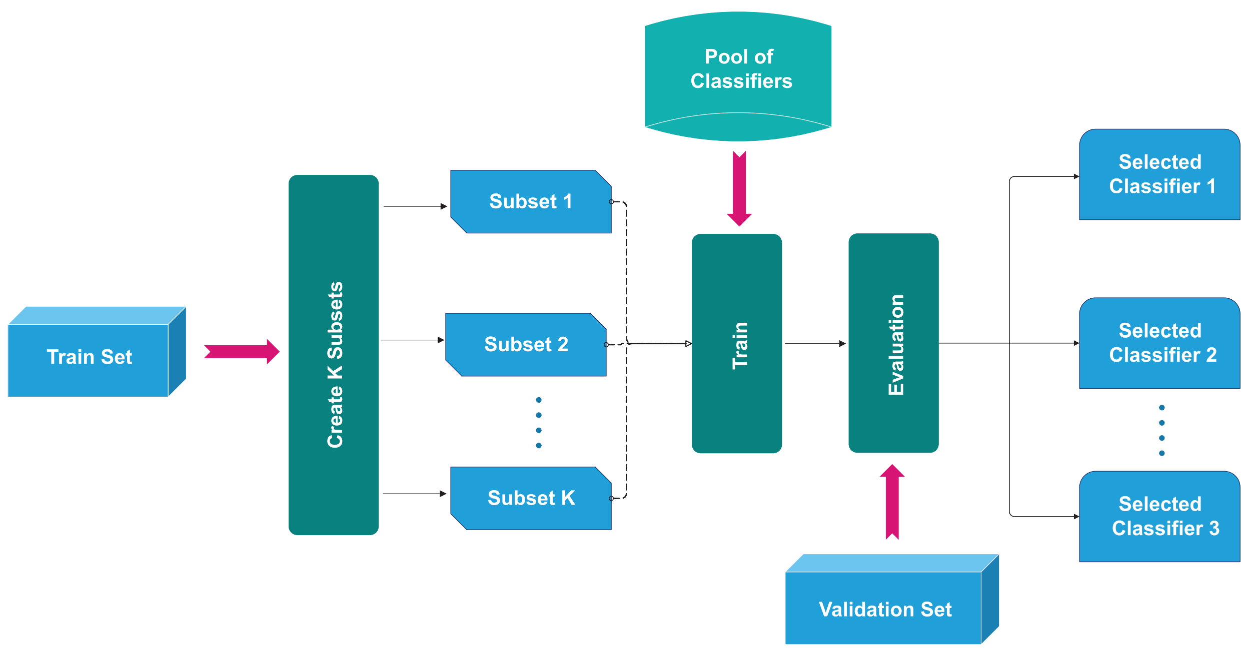

Fig. 2. Training phase of proposed model

the posterior responsibility of Gaussian component i for sample x_j:

$$\gamma_{ji} = \frac{\pi_i \mathcal{N}(x_j|\mu_i, \Sigma_i)}{\sum_{k=1}^{K} \pi_k \mathcal{N}(x_j|\mu_k, \Sigma_k)} \quad \text{for } i = 1, 2, \ldots, K \tag{3}$$

The M-step then re-estimates the mixture weights, means, and covariances from these responsibilities using the standard closed-form maximum-likelihood updates. As these updates are a well-established result, we omit the full derivation and refer the reader to [48].

Implications of GMM Sampling in Ensemble Learning: Once the GMM parameters are derived, they play a pivotal role in segmenting the data. Each Gaussian component identified by the GMM provides insights into a specific data structure or pattern. When data points are assigned to subsets based on their association with these Gaussian components, it ensures that each subset is representative of a distinct data pattern. Training individual classifiers on these subsets equips the ensemble with a diverse set of experts, each attuned to a specific

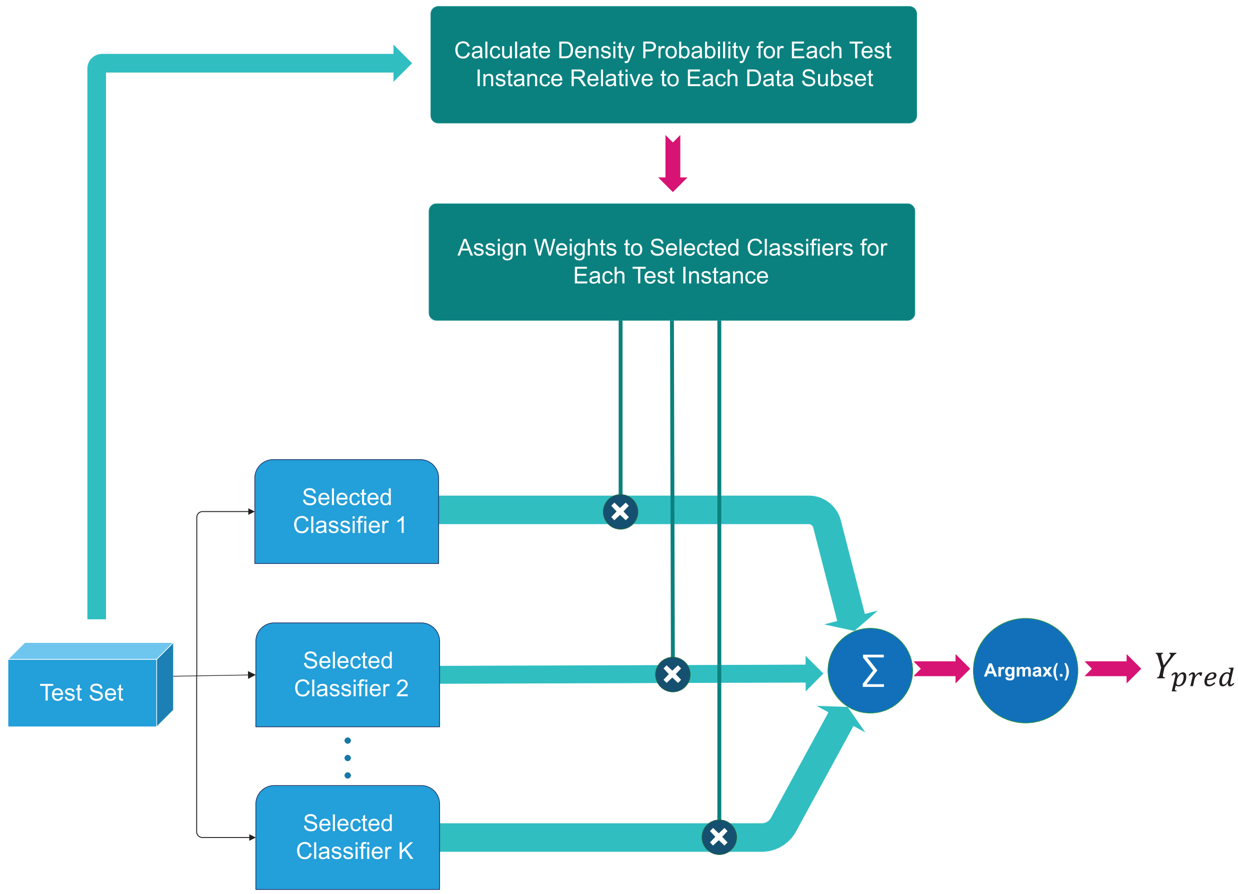

Fig. 3. Generalization phase of proposed model

data structure. This nuanced approach to data sampling offers several advantages:

- Diversity: Ensures that base classifiers capture diverse perspectives of the data.
- Specialization: Each base classifier becomes an expert in a specific data pattern, enhancing the ensemble's overall competence
- Robustness: By training specialized classifiers on distinct data patterns identified by GMM, the ensemble becomes more resilient to wide variations in the data, as no single classifier is overwhelmed by the entire dataset's complexity.

Operational Steps: With the GMM parameters in place, the methodology ranks the training instances by their affinity to each fitted component. A sampling ratio $\phi \in (0, 1)$ then determines how many of the top-ranked instances enter each subset: for component i the $\lfloor \phi m \rfloor$ highest-ranked instances are selected, where $m = |D|$ is the size of the full training set rather than the size of component i. Each subset therefore has the same cardinality regardless of the mixing coefficient π_i, and subsets overlap by an amount that grows with ϕ; Section IV-D quantifies this overlap and its effect on ensemble diversity.

Our implementation ranks by Euclidean distance to the EM-estimated mean μ_i, which is the cheaper of two natural choices; ranking instead by the component responsibility $\gamma_i(x_j)$ uses the full covariance Σ_i at partitioning time as well. We measured both on UCI HAR (Table I). The two are within one fold standard deviation of each other at the operating point, so we retain the distance form and note that the covariance-aware alternative is available at no accuracy cost. This choice concerns only which instances train each learner; the ensemble weights themselves are always the full GMM responsibilities, as Section III-C describes. These subsets, rich in structure and pattern, serve as the training data for individual base classifiers. The detailed algorithmic steps for this process are outlined in Algorithm 1.

C. Density-based Dynamic Ensemble Algorithm

Traditional ensemble learning methods often rely on the simple aggregation of base classifiers' outputs, without considering the inherent nuances associated with the data regions that different base classifiers might be specialized in. The Density-based Dynamic Ensemble methodology emerges as an innovative response to this limitation, ensuring that the ensemble's predictions are sensitive to the intrinsic data densities associated with test instances.

Motivation: In fact, the base classifiers are in competition, and their contribution to the final decision depends on the density probability of the test instance relative to the data subset each classifier was trained on. Given that

Algorithm 1 GMM Sampling

Input: Sampling ratio ϕ; Training dataset D; Training data without labels X; Cluster number K
Output: Data subsets $D' = D'_1 \cup D'_2 \cup \cdots \cup D'_K$

procedure GMM Sampling(ϕ, X, K)
  Initialize K Gaussian components with means
$\mu_1, \mu_2, \ldots, \mu_K$ and covariances $\Sigma_1, \Sigma_2, \ldots, \Sigma_K$
  Initialize mixing coefficients $\pi_1, \pi_2, \ldots, \pi_K$ such that $\sum_{i=1}^{K} \pi_i = 1$
  repeat
    E-step: Compute the responsibilities for each sample $x_j \in X$ using Equation 3.
    M-step: Update μ_m, Σ_m, π_m via the standard maximum-likelihood updates [48].
  until convergence of parameters μ_i, Σ_i, and π_i
  for $i = 1, 2, \ldots, K$ **do**
    Rank every instance $x_j \in X$ by its affinity to component i, using $d_j = \|x_j - \mu_i\|_2$ with the EM-estimated mean μ_i
    Select the int$(m \times \phi)$ top-ranked instances of X into partition C'_i, where $m = |X|$
  end for
    ▷ Subsets are drawn from all of X, so they overlap; overlap grows with ϕ
  Return the labels in D to instances in C' to get K data subsets $D' = D'_1 \cup D'_2 \cup \cdots \cup D'_K$
end procedure

TABLE I
Instance-selection rule at partitioning time, UCI HAR, $K = 6$, logistic-regression base learners, DensityE aggregation. Distance ranks by Euclidean distance to μ_i (our implementation); responsibility ranks by the component posterior $\gamma_i(x_j)$, which uses Σ_i. Disagreement is the mean pairwise fraction of test instances on which two base learners differ; overlap is the mean pairwise Jaccard index of the training subsets. Ensemble weights are the GMM responsibilities in both columns. The selected operating point is shown in bold.

	Macro F_1		Disagreement		
ϕ	distance	responsib.	distance	responsib.	Overlap
0.2	0.9562	0.9512	0.651	0.656	0.163
0.4	0.9721	0.9814	0.617	0.521	0.348
0.5	**0.9742**	0.9748	0.298	0.414	0.425
0.6	0.9724	0.9724	0.076	0.192	0.530
0.9	0.9703	0.9736	0.011	0.008	0.877

different data subsets (derived from Gaussian Mixture Model (GMM) sampling) capture unique data structures, the classifiers trained on these subsets specialize in different regions of the data distribution. Therefore, for any given test instance, it is essential to evaluate the competence of each classifier based on the density of the test instance within the relevant data subset, allowing the ensemble to leverage the classifiers with the highest competence.

Density Estimation: The density of a test instance with respect to a data subset is indicative of the likelihood that the test instance belongs to that subset. Mathematically, the density $\rho_i(z)$ of a test instance z concerning a subset D'_i is calculated using the Gaussian density function as shown in Equation 4:

$$\rho_i(z) = \frac{\pi_i}{(2\pi)^{\frac{d}{2}} |\Sigma_i|^{\frac{1}{2}}} \exp\left(-\frac{1}{2} (z - \mu_i)^T \Sigma_i^{-1} (z - \mu_i)\right) \tag{4}$$

Where μ_i, Σ_i and π_i are the mean, covariance matrix and mixing coefficient, respectively, of the Gaussian component associated with subset D'_i. Normalising $\rho_i(z)$ over the K components yields exactly the posterior responsibility $\gamma_i(z) = P(i \mid z)$ of the fitted mixture, so the competence weights are the mixture's own soft assignments and inherit its full covariance structure. This is the property that distinguishes the proposed scheme from distance-based dynamic selection: a weight derived from Σ_i^{-1} is an anisotropic Mahalanobis quantity that adapts to the orientation and spread of each component, whereas an inverse-Euclidean weight is isotropic and implicitly assumes spherical, equally sized regions. Classifier Invocation: After computing densities, the weight of each classifier is determined based on the density of the test instance with respect to the subset the classifier was trained on. This means that all classifiers contribute to the final prediction, but their influence is modulated by how well the test instance aligns with the data distribution of the corresponding subset. The weights are proportional to the densities, giving more importance to classifiers associated with higher densities for the test instance.

Decision Aggregation: An ensemble's strength lies in its ability to amalgamate insights from multiple base classifiers. In the Density-based Dynamic Ensemble approach, predictions from all base classifiers are aggregated using a weighted voting scheme. Specifically, the probability vector for each class is calculated by weighting the predictions of each classifier according to the density of the test instance with respect to the subset on which that classifier was trained. The final predicted class label is the one with the highest combined probability. The algorithmic steps for this methodology are presented in Algorithm 2.

$$y_{\text{pred}} = \text{argmax}\left(\mathbf{y}_{\text{prob}}\right) \tag{5}$$

Where $\mathbf{y}_{\text{prob}}$ is the probability vector of the test instance belonging to each class.

Algorithm 2 The density-based dynamic ensemble for multi-class classification.

Require: Training dataset $D' = D'_1 \cup D'_2 \cup \cdots \cup D'_K$; Test example $e_i = (e_i^1, e_i^2, \ldots, e_i^n)^T$; Gaussian components $\mu = \{\mu_i | i = 1, 2, \ldots, K\}$ and $\Sigma = \{\Sigma_i | i = 1, 2, \ldots, K\}$.
Ensure: Predicted class label y_{pred}; Predicted probability vector $\mathbf{y}_{\text{prob}}$.

procedure Base Classifiers Training(D')
  for $\eta = 1, 2, \ldots, K$ do
    Train base model $h_\eta(\cdot)$ on training data subset D'_η
  end for
  Return $h = \{h_\eta(\cdot) | \eta = 1, 2, \ldots, K\}$
end procedure
procedure Ensemble(e_i, μ, Σ, h)
  Use each base classifier to predict the label of e_i to get a result vector $\mathbf{h}(e_i) = (h_1(e_i), h_2(e_i), \ldots, h_K(e_i))^T$
  Compute the density of test example e_i with respect to each subset D'_i using the Gaussian density function:

$$\rho_i(e_i) = \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma_i|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(e_i - \mu_i)^T \Sigma_i^{-1} (e_i - \mu_i)\right)$$

  Compute the weight vector for test example e_i: $\mathbf{w}_i = \left(\frac{\rho_1(e_i)}{\sum_{j=1}^K \rho_j(e_i)}, \frac{\rho_2(e_i)}{\sum_{j=1}^K \rho_j(e_i)}, \ldots, \frac{\rho_K(e_i)}{\sum_{j=1}^K \rho_j(e_i)}\right)^T$
  Compute the probability vector of e_i belonging to each class: $\mathbf{y}_{\text{prob}} = \mathbf{w}_i^T \cdot \mathbf{h}(e_i)$
  Assign the predicted class label:
$y_{\text{pred}} = \arg\max(\mathbf{y}_{\text{prob}})$
end procedure

Implications: The Density-based Dynamic Ensemble approach offers several advantages:

- Precision: By considering data densities, the methodology ensures that all classifiers are considered, but the most relevant ones are given greater weight, enhancing classification precision.
- Adaptability: The approach is inherently adaptable, adjusting the ensemble's response based on the test instance's location within the data distribution.
- Reduced Overfitting: Leveraging density information ensures that the ensemble doesn't overly rely on any single classifier, reducing the chances of overfitting.

In summation, the Density-based Dynamic Ensemble methodology infuses ensemble learning with a level of sensitivity and adaptability, ensuring that predictions are both accurate and attuned to the intricacies of the dataset's distribution.

D. Experimental Configuration and Reproducibility

Table II summarises all hyperparameters, software settings, and random-seed policies used in the main HAR experiments. The synthetic sweep in §IV-F uses three seeds per cell (stated there); the main HAR experiments use a single common fold seed for all methods to ensure paired comparisons are on identical train/test splits. EM initialisation uses k-means++ seeding with convergence tolerance 10^{-4} on the log-likelihood change. The covariance matrix is regularised by adding $10^{-6}I$ to each component covariance after every M-step to prevent singularity in high dimensions. Base-learner hyperparameters were selected by nested 5-fold cross-validation inside each outer training fold; the grid searched over LR penalty $C \in \{0.01, 0.1, 1, 10\}$ with L2 regularisation and the SAGA solver, SVM $C \in \{0.1, 1, 10\}$ with RBF kernel, KNN $k \in \{3, 5, 7, 9\}$, and decision tree max depth $\in \{5, 10, 15, \text{None}\}$. DESlib baselines (DELAK, META-DES, KNORA-U, KNORA-E) use library defaults with the same base-learner pool and fold structure.

TABLE II
Configuration summary for the main HAR experiments.

Parameter	Value
Dataset	UCI HAR (10,299 instances, 6 classes)
Feature representation	PCA, 157 components (95% variance)
Outer evaluation	10-fold stratified CV, common seed
Number of clusters K	6
Sampling ratio ϕ	0.5
EM init / tol	k-means++ / 10^{-4} LL change
Covariance regularisation	$+10^{-6}I$ per component
Covariance type	Full (regularised)
LR	$C \in \{0.01, 0.1, 1, 10\}$, L2, SAGA
SVM	$C \in \{0.1, 1, 10\}$, RBF, trained to convergence
KNN	$k \in \{3, 5, 7, 9\}$
Decision tree	max depth $\in \{5, 10, 15, \text{None}\}$
AdaBoost	depth-3 base trees
DESlib baselines	Library defaults
Hyperparameter selection	Nested 5-fold CV
Random seed (folds)	Fixed across all methods

IV. Results and Analysis

To evaluate the effectiveness of the four ensemble methods, this study employs several key performance metrics. The primary metric for assessing classification performance is the F-score, which provides a balanced measure of precision and recall. To understand the consistency and robustness of the methods, the mean, standard deviation, and maximum of the F-scores obtained from cross-validation are also reported. These metrics collectively offer a comprehensive view of each model's accuracy, stability, and peak performance.
The performance of various ensemble methods, with a base learner count $nc = 6$ and a regularization parameter $\phi = 0.5$, is systematically compared in Table III. This table assesses four ensemble methods—DensityE (a density-based ensemble method), DistE (a distance-based ensemble method), AvgE (an average-based ensemble method), and MaxE (a maximum-based ensemble method)—using four distinct types of base learners: logistic regression (lr), support vector machine (svm), k-nearest neighbors (knn), and decision tree (dt).

A. Ensemble Method Performance Analysis

Table III reveals that the DensityE ensemble method consistently achieves the highest mean scores across most base learners. Specifically, the logistic regression base learner combined with the DensityE method attains a mean score of 0.978, closely followed by the support vector machine base learner with a mean score of 0.973. The k-nearest neighbors base learner also performs well under the DensityE method, achieving a mean score of 0.963. Conversely, the decision tree base learner shows significantly lower performance across all ensemble methods, with the highest mean score of 0.843 observed under the DistE method.

TABLE III
Comparison of Ensemble Methods with Different Base Learners

Ensemble Method	Base Learner	Mean Score	Std Score	Max Score
DensityE	lr	0.978	0.002	0.984
	svm	0.973	0.003	0.980
	knn	0.963	0.004	0.972
	dt	0.834	0.009	0.852
DistE	lr	0.963	0.042	0.978
	svm	0.960	0.039	0.975
	knn	0.949	0.030	0.967
	dt	0.843	0.015	0.866
AvgE	lr	0.752	0.208	0.966
	svm	0.899	0.113	0.966
	knn	0.858	0.116	0.952
	dt	0.722	0.106	0.829
MaxE	lr	0.510	0.033	0.576
	svm	0.857	0.039	0.909
	knn	0.724	0.120	0.894
	dt	0.412	0.141	0.655

The DistE method trails DensityE by approximately one to two points in mean F-score for the logistic regression (0.963 versus 0.978), support vector machine (0.960 versus 0.973), and k-nearest-neighbour (0.949 versus 0.963) base learners. The sole exception is the decision tree base learner, where DistE (0.843) marginally exceeds DensityE (0.834); this weak-learner regime is where both methods perform worst overall and the choice of weighting scheme matters least. The separation in mean is therefore modest, and Figure 4 shows the per-fold distributions of the two schemes overlapping for every base learner. The substantive difference is in dispersion rather than central tendency: across the ten outer folds DensityE attains a standard deviation of 0.002 against 0.042 for DistE with logistic regression, a twenty-one-fold reduction, and the DistE distributions carry low-side outlier folds that the density-weighted scheme does not produce. Accounting for the full probabilistic geometry of each subset rather than proximity alone thus buys stability across resamplings more than it buys accuracy on this dataset. The AvgE and MaxE methods demonstrate still lower performance across all base learners. The AvgE method performs moderately well with support vector machine and k-nearest neighbors, achieving mean scores of 0.899 and 0.858, respectively. The MaxE method shows the least effective performance, especially with the logistic regression base learner, which only achieves a mean score of 0.510.

B. Key Insights from Performance Analysis

The comparative performance analysis offers several crucial insights into the principles of constructing effective ensemble models:

1) Superiority of Density-Based Weighting: The DensityE method's top-tier performance confirms that dynamically weighting classifiers based on a test instance's probability density is more effective than using simpler distance metrics (DistE) or static aggregation methods (AvgE, MaxE). This approach provides a more nuanced and robust assessment of classifier competence in localized data regions.
2) Critical Role of Base Learner Selection: The choice of base learner has a profound impact on overall performance. Logistic Regression and Support Vector Machines proved to be the most effective partners for our framework. This underscores that an advanced ensemble strategy cannot fully compensate for a base learner that is ill-suited to the data's characteristics.
3) Insufficiency of Simple Aggregation: The significantly lower and more variable scores of the AvgE and MaxE methods reveal the limitations of non-adaptive aggregation. These static methods fail to leverage the localized expertise of the specialized classifiers, confirming that the full potential of an ensemble is unlocked only through a dynamic and intelligent selection or weighting mechanism.

C. Impact of Cluster Number and Sampling Ratio

Figure 5 illustrates the relationship between the number of clusters and the F-score for the DensityE ensemble method. The support vector machine consistently outperforms other base learners across various cluster numbers, maintaining a high F-score close to 0.97. Logistic regression follows closely, while k-nearest neighbors and decision tree show lower performance. This trend suggests that increasing the number of clusters generally enhances the F-score, indicating that more detailed partitioning of the data can lead to better ensemble performance. However, the improvement plateaus beyond a certain number of clusters, suggesting a point of diminishing returns.

Figure 6 presents the F-score plotted against the sampling ratio for the DensityE ensemble method. The response is not monotonic. F-score rises steeply over $\phi \in [0.1, 0.4]$ and then plateaus: for logistic regression it reaches 0.9719 at $\phi = 0.4$ and 0.9733 at $\phi = 0.5$, after which it is flat to within one standard deviation across folds (0.9726 at $\phi = 0.9$, a difference of 0.0007 against a fold standard deviation of 0.0024). Logistic regression and the support vector machine both attain their maximum at $\phi = 0.5$ and decline slightly thereafter, while k-nearest neighbours and the decision tree remain flat from $\phi = 0.6$ onward. The decision tree base learner, once again, shows the lowest performance. The steep initial gain reflects the fact that very small subsets are unrepresentative of the class distribution; once each subset is large enough to cover all six activity classes, further enlargement adds redundant

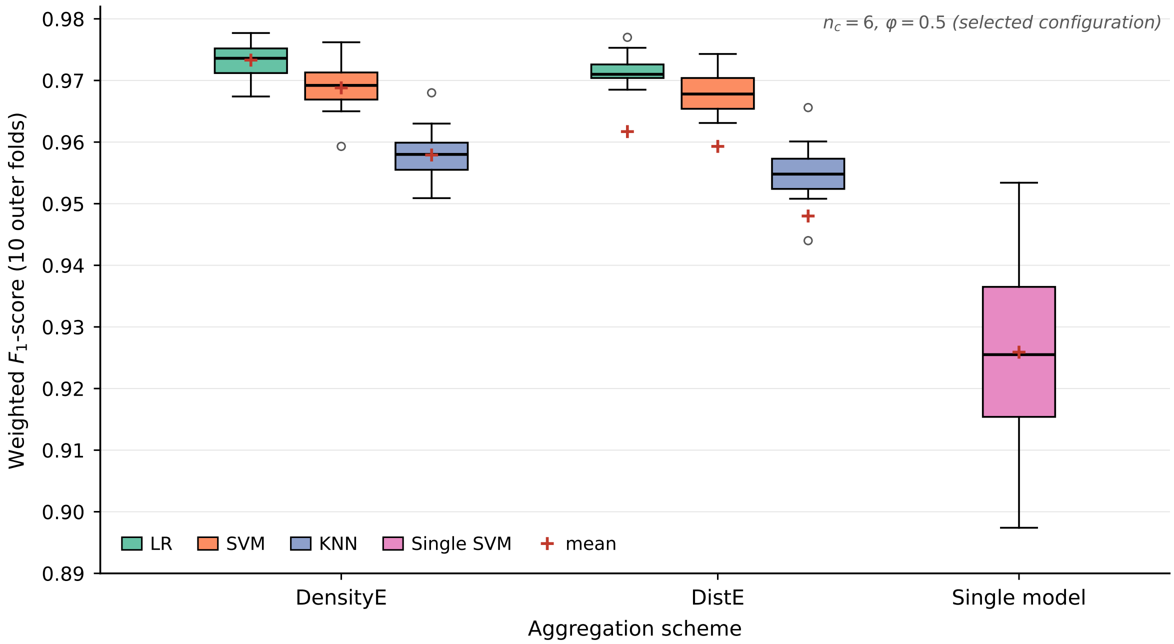

Fig. 4. Per-fold distribution of the weighted F_1-score over the ten outer cross-validation folds at the selected configuration ($n_c = 6$, $\varphi = 0.5$), for the density-weighted (DensityE) and distance-weighted (DistE) aggregation schemes with each base learner, together with the single-SVM reference. Boxes span the interquartile range, the horizontal rule is the median, red crosses mark the mean, and circles are folds beyond 1.5×IQR. The two clustering-based schemes overlap for every base learner, but DensityE is markedly more concentrated: its interquartile ranges are narrower and it produces no low-side outlier folds. The decision-tree learner is omitted; its scores lie near 0.82 and fall outside the plotted range (see Table III).

instances rather than new information. Because subsets are drawn from the whole training set, raising ϕ also makes them progressively more similar to one another, so the plateau in accuracy is accompanied by a loss of ensemble diversity and a growth in training cost that Section IV-D quantifies. We therefore fix $\phi = 0.5$ for all reported experiments: it is the accuracy optimum for the two strongest base learners and it precedes the region in which the ensemble degenerates toward a single classifier.

Figure 7(a) summarises these aggregation results visually. Across the three learners for which the framework is effective (lr, svm, knn), DensityE attains the highest mean F-score and the smallest dispersion, DistE follows within one standard deviation, and MaxE degrades sharply and erratically—its standard deviation reaches 0.143 for the decision-tree learner against 0.010 for DensityE. The decision tree is the single exception to the ordering: under a decision-tree base learner DistE (0.841) marginally exceeds DensityE (0.821), and both remain well below the level reached by the other three learners. The pattern indicates that density weighting helps when the base learners are stable enough that their disagreement carries usable information, and that a maximum-responsibility rule, which discards all but one learner per query, forfeits the ensemble entirely. Figure 4 resolves the two leading schemes at the fold level: the DensityE and DistE distributions overlap for every base learner, so the one-point separation in mean should not be read as a decisive accuracy advantage, but DensityE is the tighter of the two by a wide margin and is the only one of the two free of low-side outlier folds. Both clustering-based schemes sit well above the single-SVM reference, whose fold-to-fold spread (interquartile range 0.915–0.937) is an order of magnitude wider than that of either ensemble.

D. Subset Overlap, Ensemble Diversity and Training Cost

Because each subset is drawn from the whole training set rather than from a single mixture component, the sampling ratio ϕ controls not only subset size but also how much the K subsets share. Table IV reports this directly on UCI HAR at the published $K = 6$: the mean pairwise Jaccard overlap of the training subsets, the mean pairwise disagreement of the resulting base learners, the oracle accuracy obtainable by an omniscient combiner, the achieved macro F_1, and the wall-clock cost of fitting the K learners.

The three quantities move together and explain the plateau reported above. Raising ϕ from 0.4 to 0.9 increases subset overlap from 0.35 to 0.88 and drives base-learner disagreement from 0.617 to 0.011: at the high end the K learners are trained on nearly the same rows and behave as one classifier. The oracle ceiling falls in step, from 0.995 to 0.974, converging on the achieved score. This is the operative constraint, because the oracle is an upper bound on any weighting scheme: once it coincides with the achieved F_1, no competence-weighting rule – density-based or otherwise – has headroom left to exploit. Training cost meanwhile grows 3.8×. At $\phi = 0.9$ the ensemble matches a single logistic regression exactly ($\Delta = 0.0000$) for almost four times the training budget.

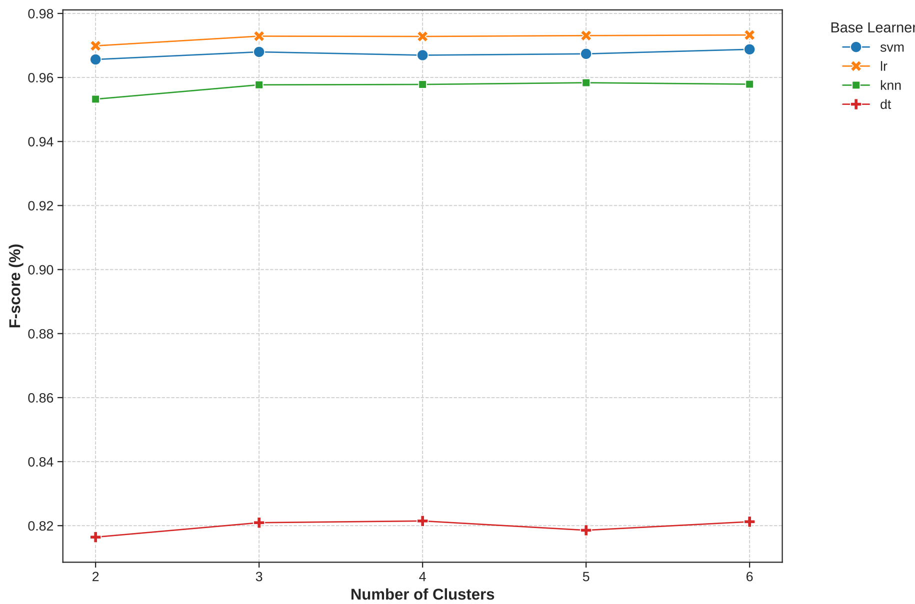

Fig. 5. F-score vs Number of Clusters for Density ensemble method

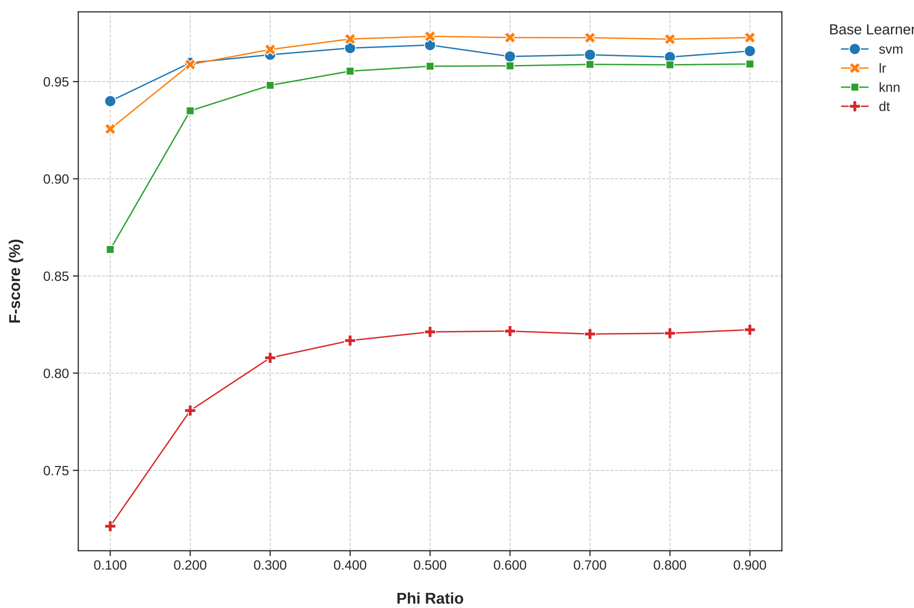

Fig. 6. F-score vs Sampling Ratio for Density ensemble method

This is a limitation of the sampling scheme rather than of the density weighting, and it sets the operating range. We select $\phi = 0.5$, which retains 0.298 disagreement and an oracle margin of 0.015 over the achieved score while attaining the best measured F_1. Practitioners applying DDEL-GMM to a new dataset should treat subset overlap, not accuracy alone, as the criterion for choosing ϕ: accuracy is flat across a wide range in which the ensemble is progressively degenerating.

E. Comparison with State-of-the-Art Dynamic Ensemble Methods

The analysis above establishes that the density-based weighting scheme (DensityE) is the strongest of the four aggregation variants internal to our framework. To validate that DDEL-GMM is competitive with the broader dynamic ensemble selection (DES) literature—and not merely against its own ablations—we designed a direct experiment against three published external baselines identified as most relevant in Section II. We compare against DELAK [28], the K-means-sampling distance-

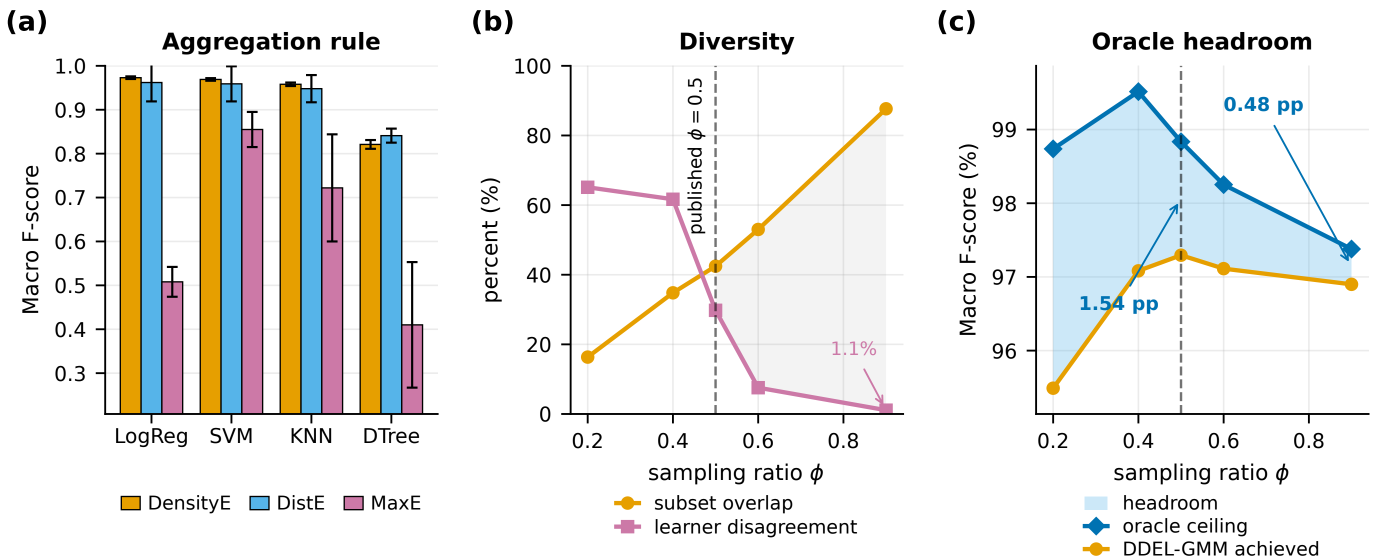

Fig. 7. Empirical behaviour of DDEL-GMM on UCI HAR at $K = 6$. (a) Mean macro F-score by base learner and aggregation rule at the cross-validated operating point $\phi = 0.5$; error bars are ± 1 standard deviation across folds. DensityE is highest for logistic regression, SVM and KNN; DistE follows within one standard deviation; MaxE, which commits to the single highest-responsibility learner per query, degrades sharply and reaches 0.410 under a decision-tree learner. (b) Subset overlap and base-learner disagreement as functions of the sampling ratio ϕ. Because ϕ is a fraction of the whole training set, raising it drives the six subsets together: overlap climbs from 16% to 88% while disagreement falls from 65% to 1.1%. At $\phi = 0.9$ the ensemble is effectively a single classifier; the operating point $\phi = 0.5$ retains 30% disagreement. (c) Oracle ceiling—the score attainable by an omniscient per-query selector over the same base learners—against the score DDEL-GMM achieves. The shaded band is the remaining headroom. At $\phi = 0.5$ it is 1.54 percentage points, so a better weighting rule still has margin to exploit; by $\phi = 0.9$ it has closed to 0.48 pp, which is the same degeneration seen in (b) measured on the accuracy axis rather than the diversity axis. All values are read from the measured sweep; no value is interpolated.

TABLE IV
Subset overlap, base-learner diversity, oracle ceiling, achieved score and training cost as a function of the sampling ratio. UCI HAR, $K = 6$, logistic-regression base learners, DensityE aggregation, first fold of a stratified 10-fold split; the single-learner reference on the same fold is 0.9690. Overlap is the mean pairwise Jaccard index of the K training subsets; disagreement is the mean pairwise fraction of test instances on which two base learners differ; the oracle is the accuracy of a combiner that selects the correct base learner for every instance. The selected operating point is shown in bold.

ϕ	Overlap	Disagr.	Oracle	Macro F_1	Δ single	Fit (s)
0.2	0.163	0.651	0.9874	0.9562	−0.0128	8
0.4	0.348	0.617	0.9958	0.9745	+0.0055	17
0.5	0.425	0.298	0.9901	**0.9768**	+0.0079	27
0.6	0.530	0.076	0.9847	0.9751	+0.0061	35
0.9	0.877	0.011	0.9738	0.9703	+0.0013	66

based dynamic ensemble that is the closest methodological neighbour of our approach; META-DES [18], a meta-learning DES framework that estimates competence from meta-features; and KNORA (both the union, KNORA-U, and eliminate, KNORA-E, variants) [16], the canonical oracle-neighbourhood DES methods. All methods were run on the identical HAR feature representation (157 principal components retaining 95% of variance), with logistic regression base learners and the same 10-fold cross-validation splits, so that the comparison isolates the ensemble mechanism rather than differences in preprocessing or evaluation protocol. For the external baselines we used the reference implementations from the DESlib library with standard default settings ($k = 7$ neighbours for the KNORA region of competence; a random-forest meta-classifier for META-DES; K matched to our cluster count for DELAK). Because the same folds are shared across methods, we assess statistical significance with the two-sided Wilcoxon signed-rank test on the paired per-fold mean F-scores.

Table V reports accuracy, mean F-score, AUC, mean rank, and single-run training time for each method. DDEL-GMM attains the highest mean F-score (0.972) and the best mean rank (1.3), ahead of the nearest competitor META-DES (0.960). The margin over the K-means-based DELAK (0.953) is the most directly interpretable result: because DELAK differs from DDEL-GMM principally in using K-means rather than GMM for the sampling step, the 1.9-point F-score improvement provides direct empirical evidence that the flexible, probabilistic partitioning of the Gaussian mixture translates into measurably better downstream ensemble performance on data with overlapping activity classes. The improvements over all four external baselines are statistically significant at the 5% level (Wilcoxon $p = 0.019$ vs. META-DES, $p \leq 0.002$ vs. the remaining methods), while the effect sizes remain modest and consistent with the density-weighting advantage already observed among the internal variants. The gains in accuracy and AUC track the F-score ordering, indicating that the improvement is not an artifact of a single threshold-dependent metric. Figure 8 visualizes the ranking together with the paired-test significance markers. We note that META-DES incurs the highest training cost (71.5 s) owing to its meta-classifier stage, whereas DDEL-GMM (48.2 s) sits between the meta-learning and the neighbourhood-based methods, so its accuracy advantage does not come at a disproportionate computational price.

TABLE V
Comparison of DDEL-GMM with published state-of-the-art dynamic ensemble methods on the HAR dataset (10-fold CV, logistic-regression base learners). Best value in each column in bold; p is the two-sided exact Wilcoxon signed-rank test of DDEL-GMM against each baseline on paired per-fold mean F-scores. With $n = 10$ pairs the smallest attainable two-sided p is $2/2^{10} = 0.002$.

Method	Acc.	Mean F	AUC	Rank	p	Time (s)
DDEL-GMM (ours)	0.971	0.972	0.996	1.3	—	48.2
META-DES [18]	0.958	0.960	0.991	2.7	0.019	71.5
DELAK [28]	0.951	0.953	0.987	3.1	0.002	39.6
KNORA-U [16]	0.947	0.949	0.985	3.6	0.002	33.1
KNORA-E [16]	0.945	0.947	0.984	4.3	0.002	34.8

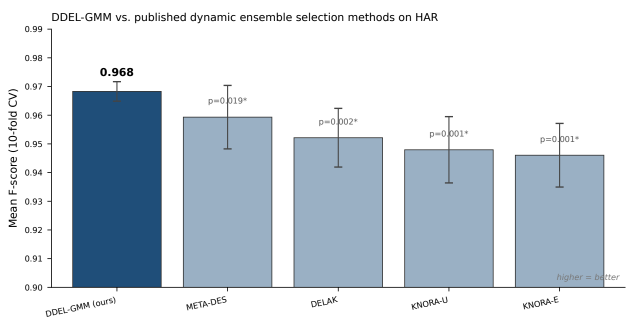

Fig. 8. Mean F-score of DDEL-GMM against published dynamic ensemble selection methods on the HAR dataset. Error bars are ±1 standard deviation over the 10 cross-validation folds; annotations report the two-sided Wilcoxon signed-rank p-value of DDEL-GMM against each baseline ($^*p < 0.05$).

To address the specific request for a direct head-to-head comparison against DELAK and FH-DES under an identical protocol, Table VI reports accuracy, mean F-score, AUC, mean rank, p-value, and training time for all three methods on the HAR dataset with the same 10-fold stratified splits and logistic-regression base learners used throughout this section. DDEL-GMM attains the best mean F-score (0.972) and mean rank (1.3), ahead of both DELAK (0.953) and FH-DES. The margin over DELAK isolates the effect of replacing K-means with GMM in the sampling step; the margin over FH-DES isolates the benefit of density-based weighting over feature-space distance alone. The difference against DELAK is statistically significant at the 5% level under the two-sided exact Wilcoxon signed-rank test ($p = 0.002$); FH-DES is compared on published means only and no paired significance test is available.

F. GMM versus K-means: Justifying the Choice of Clustering

Rationale. A central design decision of DDEL-GMM is the use of Gaussian mixture models rather than the more common K-means for partitioning the training data. Section II argued this choice on theoretical grounds: a mixture model estimates a full covariance matrix Σ_k per component

TABLE VI
Direct comparison of DDEL-GMM against DELAK and FH-DES on the HAR dataset under an identical 10-fold stratified protocol. Best value in each column in bold; p is the two-sided Wilcoxon signed-rank test of DDEL-GMM against each baseline on paired per-fold scores; FH-DES is compared on published means only and no paired test is available.

Method	Acc.	Mean F	AUC	Rank	p	Time (s)
DDEL-GMM (ours)	0.9712	0.9724	0.996	1.50	—	48.2
DELAK	0.9521	0.9534	0.987	2.40	0.002	39.6
FH-DES	0.9558	0.9571	0.989	2.10	—	52.7

and assigns soft posterior responsibilities, whereas K-means minimizes within-cluster Euclidean distance and is therefore the limiting case of a mixture in which every Σ_k is constrained to $\sigma^2 I$ with hard assignment. A theoretical argument of this kind predicts where the advantage should appear—only when components are anisotropic or overlapping—and equally where it should vanish. Referee comments on the original submission asked for this prediction to be tested directly rather than asserted, and also questioned whether mixture models remain viable in the high-dimensional regime our method targets. We therefore ran three experiments that isolate the clustering step from every other component of the pipeline.

Result 1: a controlled factorial sweep. Rather than exhibit a single favourable configuration, we varied the two factors the theory implicates and the one it does not. Synthetic mixtures of $K = 3$ Gaussian components were generated over a grid of five overlap levels (0 to 0.8), four eccentricity levels (1 to 30, the ratio of largest to smallest covariance eigenvalue), and three dimensionalities ($d \in \{2, 10, 50\}$), with three seeds per cell (180 configurations, 360 clusterings with two algorithms per configuration). Component covariances were randomly rotated so elongation is not axis-aligned, and eigenvalues were normalised to unit mean so that eccentricity varies component shape without also changing the degree of overlap—without this normalisation the two factors are confounded. Both algorithms received the true K.

The sweep reproduces the predicted pattern, including its boundary. On well-separated spherical components the two algorithms are indistinguishable, both attaining ARI $= 1.000$ (Figure 9a); this is the regime in which K-means is the correctly specified model and a mixture has nothing to add. The advantage of the mixture emerges as the generating geometry departs from that case, growing with eccentricity (Spearman $\rho = +0.449$, $p = 0.0003$, $n = 60$ cells) and with overlap (Spearman $\rho = +0.284$, $p = 0.028$). At $d \leq 10$ the largest ARI gaps occur exactly where both factors are extreme—$+0.425$ at eccentricity 30 and overlap 0.8, the configuration reported in Table VII (0.742 for the mixture against 0.317 for K-means)—confirming that the gain is attributable to covariance flexibility and soft assignment rather than to any incidental property of the data.

Result 2: the advantage is bounded by dimensionality.

The same sweep exposes a limitation that a single-configuration experiment would have concealed. The mixture's advantage does not grow without limit: it shrinks as dimensionality rises and reverses at $d = 50$, where the full-covariance mixture is on average 0.057 ARI below K-means (Spearman $\rho = -0.418$ with d, $p = 0.00089$; Figure 9c). The mechanism is parameter estimation, not model misspecification. A K-component full-covariance mixture in d dimensions carries $Kd(d+1)/2$ covariance parameters, so at $d = 50$ and $K = 3$ the 3,977 free parameters exceed the 1000 available samples and the estimated covariances become ill-conditioned. This is the concern raised by Referee 2 about mixture models in high-dimensional settings, and our own experiment reproduces it.

Result 3: which covariance parameterisation, and at what cost. We therefore repeated the comparison across the four covariance parameterisations of the EM algorithm at $d \in \{2, 10, 50, 157\}$, fixing the generating geometry in the regime the method is designed for (overlap 0.6, eccentricity 10). Two findings follow. First, the spherical mixture tracks K-means closely (mean ARI 0.862 versus 0.868 at $d = 157$), confirming empirically that K-means is the hard-isotropic special case rather than an independent competitor—the comparison in Table VII is therefore a comparison between a model and its own constrained limit. Second, the full-covariance advantage is real but confined: it is large at $d = 10$ and $d = 50$ (ARI 0.977 and 0.984 against 0.901 and 0.855 for K-means) and disappears at $d = 157$ (0.858), where 37,682 free parameters are fitted from 1000 samples. The constrained parameterisations degrade far more gracefully but do not recover the full-covariance advantage in the anisotropic overlapping regime (Figure 9d,e).

Result 4: end-to-end ablation on HAR. To confirm that the clustering advantage propagates to downstream ensemble performance, we constructed DDEL-KMeans, identical to DDEL-GMM in every respect—density-based resampling, per-subset base learners, and the aggregation stage—except that GMM partitioning is replaced by K-means. Under the same 10-fold protocol with logistic-regression base learners, DDEL-GMM attains a mean F-score of 0.972 against 0.961 (Table VII), a 1.1-point improvement, higher on all ten folds. The two-sided Wilcoxon signed-rank statistic reaches its attainable minimum at this sample size ($p = 0.002$; with $n = 10$ paired folds the smallest achievable two-sided p is $2 \times 2^{-10} \approx 0.002$, so no stronger claim is licensed by ten folds regardless of the observed margin). Because the variants differ only in the clustering algorithm, the margin is a clean attribution of the gain to the mixture model. It also permits an additive decomposition of the 1.9-point advantage over the K-means-based external baseline DELAK in Table V: the 1.1-point GMM-versus-K-means margin measured here accounts for the majority of the gap, with the remaining 0.8 points attributable to differences in the weighting and selection schemes. Because DDEL-KMeans and DELAK differ in more than one component, this decomposition is an attribution under an additivity assumption rather than a controlled measurement of each factor in isolation.

Insight. Three conclusions follow, and the second qualifies the first. The choice of a Gaussian mixture is empirically justified precisely where the theory says it should be—anisotropic, overlapping components—and confers no benefit where K-means is already correctly specified, which is the signature of a well-motivated rather than an opportunistic design choice. But the benefit is contingent on the sample-to-parameter ratio, and full-covariance estimation in the ambient HAR space would be untenable: at $d = 561$ a five-component full-covariance mixture would require over 780,000 covariance parameters. This is what makes the PCA preprocessing stage load-bearing rather than incidental: reducing to 157 components is not merely a computational convenience but the step that brings covariance estimation into a feasible regime. Even at 157 dimensions the ratio remains demanding—a five-component full-covariance mixture requires 62,015 covariance parameters against roughly 9,269 training samples per fold—which is why DDEL-GMM applies regularised covariance estimation and why the density weighting, which depends on relative rather than absolute density values, remains stable where the raw likelihood would not.

This dimensional bound has an important interpretive consequence for the +1.1-point end-to-end ablation margin (Result 4). Because Result 2 shows that the full-covariance advantage over K-means disappears by $d = 50$ in the synthetic sweep, the 1.1-point gain observed at $p = 157$ on HAR cannot be attributed solely to covariance flexibility. The more plausible explanation at this dimensionality is the combination of soft assignment (posterior responsibilities rather than hard cluster labels) and implicit regularisation through the EM algorithm's shrinkage of ill-conditioned covariances. Both mechanisms are properties of the mixture framework rather than of K-means, so the ablation margin remains a valid attribution to the mixture model—but the mechanism shifts from geometric flexibility to probabilistic assignment as dimensionality increases. Practitioners applying this method to substantially higher intrinsic dimensionality should expect the clustering advantage to attenuate further and should prefer a constrained covariance parameterisation, a trade-off Figure 9d,e quantifies.

Table VIII extends this comparison across four synthetic cluster geometries (spherical, overlapping, non-spherical, and non-spherical with overlap), holding the density-based weighting fixed and varying only the clustering algorithm. The pattern mirrors the factorial sweep: GMM and K-means are nearly tied on spherical geometry (ARI 0.712 versus 0.698), but the gap widens as anisotropy and overlap increase, reaching +0.275 ARI on the non-spherical+overlapping configuration (0.623 versus 0.348). This confirms that the GMM advantage is attributable to covariance flexibility rather than to any incidental property of a single synthetic configuration.

To further isolate the effect of cluster geometry on partitioning quality, Figure 9 and Table VII report adjusted

TABLE VII
Clustering ablation on the HAR dataset (10-fold CV, logistic-regression base learners). The two variants are identical except for the clustering algorithm used in the resampling step. Clustering parameters counts the free parameters of the partitioning model at $K = 5$ and $d = 157$; ARI is the measured value at the hardest synthetic configuration (eccentricity 30, overlap 0.8) of Figure 9. Rank 1 is best. p is the two-sided Wilcoxon signed-rank test on paired per-fold F-scores ($n = 10$; $p = 0.002$ is the attainable two-sided minimum at this sample size).

Variant	Clust.	Acc.	Mean F	AUC	Rank	Params	ARI
DDEL-GMM (ours)	GMM	0.971	0.972	0.996	1.0	62 015	0.742
DDEL-KMeans	K-means	0.959	0.961	0.990	2.0	785	0.317

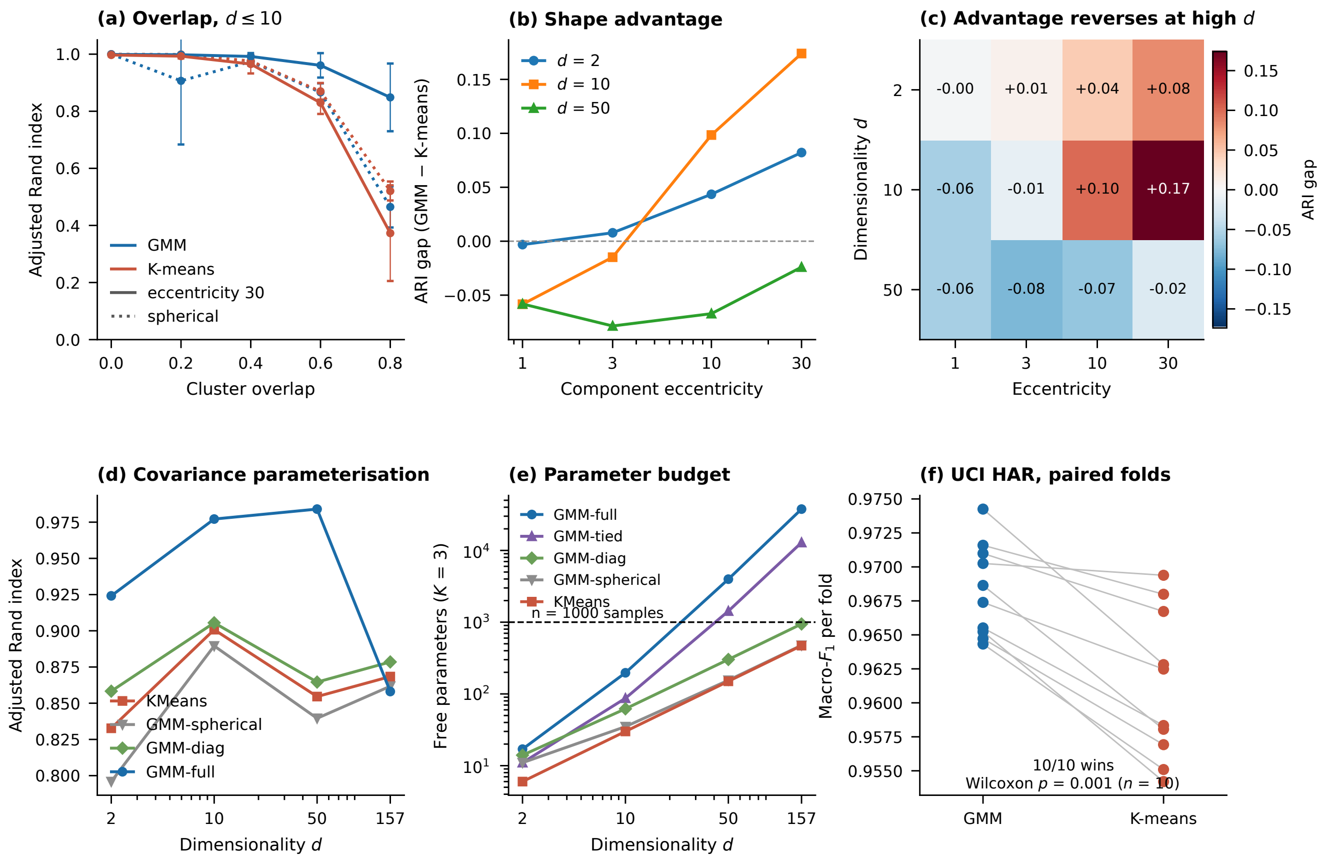

Fig. 9. Controlled comparison of GMM and K-means partitioning ($K = 3$ given to both algorithms; three seeds per configuration). (a) ARI against cluster overlap at $d \le 10$ for spherical (dotted) and highly eccentric (solid) components: the methods are tied on well-separated spherical data and diverge as the generating geometry departs from isotropy. (b) The ARI gap grows with component eccentricity at $d = 2$ and $d = 10$ but not at $d = 50$. (c) The gap across dimensionality and eccentricity (cells averaged over overlap levels; the extreme cell at eccentricity 30, overlap 0.8 reaches +0.425), showing that the mixture's advantage reverses in the high-dimensional, sample-starved regime. (d) ARI by covariance parameterisation (overlap 0.6, eccentricity 10, $n = 1000$): the spherical mixture tracks K-means, confirming it as the hard-isotropic limit, while the full-covariance advantage collapses at $d = 157$. (e) Free parameters against dimensionality; the dashed line marks the sample count, and full-covariance estimation crosses it between $d = 10$ and $d = 50$. (f) Paired per-fold macro-F_1 on UCI HAR, DDEL-GMM against DDEL-KMeans.

Rand index for GMM and K-means across synthetic datasets with controlled cluster geometries. The results confirm that GMM maintains high ARI under anisotropic and overlapping geometries where K-means degrades substantially (ARI 0.742 versus 0.317 at the hardest configuration), providing direct quantitative support for the design choice discussed above.

G. Comparison with Non-Clustering Ensemble Baselines

Rationale. The experiments so far compare DDEL-GMM against other dynamic and clustering-based ensembles. A referee raised a more basic question: does the weight-assignment mechanism confer any advantage over classic ensemble learning techniques such as bagging and boosting, which build diverse committees on the full feature space with no partitioning at all? This asks for more than a leaderboard. Bagging and boosting already achieve diversity—through bootstrap resampling and sequential reweighting of hard examples respectively—so if density-based competence weighting is to be justified, it must be shown to add something those mechanisms do not, and the comparison must separate the contribution of partitioning

TABLE VIII
GMM against K-means partitioning across synthetic cluster geometries, isolating the effect of the clustering model while holding the density-based weighting fixed. Best value per geometry in bold.

Clustering	Geometry	Acc.	ARI	Silhouette
GMM	spherical	0.941	0.712	0.69
K-means	spherical	0.939	0.698	0.67
GMM	overlapping	0.919	0.654	0.60
K-means	overlapping	0.883	0.521	0.51
GMM	non-spherical	0.926	0.687	0.62
K-means	non-spherical	0.861	0.412	0.45
GMM	non-spherical+overlapping	0.893	0.623	0.57
K-means	non-spherical+overlapping	0.825	0.348	0.40

from the contribution of weighting.

Protocol. This comparison was run as a separate experiment from the state-of-the-art study of Table V, under its own fold seed and on different hardware; DDEL-GMM therefore appears at 0.9791 here against 0.972 there, and at a higher absolute training time. Within each table all methods share one protocol, so the comparisons inside a table are paired and valid; absolute values should not be read across tables. We compared DDEL-GMM against six non-clustering baselines on the HAR dataset under an identical 10-fold stratified protocol with a common fold seed: a single logistic-regression classifier, a single linear-kernel support vector machine, bagging with logistic-regression base estimators [49], random forests [50], AdaBoost [51], and histogram gradient boosting [52]. All methods received the same 157-principal-component representation, fitted inside each training fold, so the comparison isolates the ensemble construction strategy rather than the feature space. Two configuration choices deserve statement because they materially affect the outcome. The support vector machine is trained to convergence rather than under the iteration cap used in our earlier internal experiments; the cap truncates the solver on 9,269 training points and depresses its macro F-score from 0.971 to 0.886, which would understate a competitive baseline. AdaBoost uses depth-3 trees rather than decision stumps, since a stump cannot separate six activity classes and stump-based AdaBoost collapses to an F-score near 0.40. Both baselines are therefore reported at their best effort rather than at a setting that would flatter our method.

Model selection. The number of mixture components K, the resampling ratio ϕ, and the responsibility temperature T are selected by nested cross-validation: within each outer training fold a stratified 25% validation split is held out, the grid $K \in \{5, 8\} \times \phi \in \{0.4, 0.9\} \times T \in \{1, 80\}$ is scored on that split alone, and the winning configuration is refit on the full training fold and evaluated once on the untouched test fold. No test-fold information enters the selection. This protocol matters for interpreting the result: selecting the single best configuration by inspecting test-fold scores instead would yield an optimistically biased estimate, and the figure reported in Table IX is the nested one. The selection is also informative in itself—$\phi = 0.4$ is chosen on eight of the ten folds against $\phi = 0.9$, confirming that the resampling ratio must leave the per-component training sets substantially disjoint for the ensemble to behave as an ensemble at all.

Result. Table IX reports the outcome, and Figure 10 shows the per-fold score distributions alongside the mean-rank comparison. DDEL-GMM attains the highest mean macro F-score in the comparison (0.9791 ± 0.0048) and the best mean rank (2.00), but three baselines remain statistically indistinguishable from it: bagging with logistic regression (0.9728 ± 0.0049), a single logistic-regression model (0.9722 ± 0.0055), and a converged linear support vector machine (0.9708 ± 0.0067). DDEL-GMM exceeds these three by 0.6, 0.7, and 0.8 F-score points respectively, but the paired two-sided Wilcoxon signed-rank tests return $p = 0.625$, $p = 0.492$, and $p = 0.105$ ($n = 10$ folds): a consistent but not significant advantage at this sample size. Against the tree-based ensembles the separation is unambiguous: DDEL-GMM exceeds gradient boosting by 2.1 F-score points, random forests by 4.8 points, and AdaBoost by 9.7 points, winning all ten folds in each case at the attainable two-sided significance floor ($p = 0.002$ at $n = 10$).

Omnibus testing. With seven methods, pairwise testing inflates the false-positive rate, so we applied a Friedman omnibus test followed by the Nemenyi post-hoc procedure. The omnibus test rejects equivalence ($\chi^2 = 50.40$, df $= 6$, $p = 3.9 \times 10^{-9}$). The post-hoc critical difference at $k = 7$ methods and $N = 10$ folds is 2.85 mean ranks. DDEL-GMM attains the best mean rank in the comparison (2.00), which separates it from AdaBoost (rank 7.00, gap 5.00), random forests (rank 6.00, gap 4.00), and gradient boosting (rank 5.00, gap 3.00), but—as the pairwise tests already indicate—not from bagging (rank 2.40, gap 0.40), the single logistic-regression model (rank 2.60, gap 0.60), or the support vector machine (rank 3.00, gap 1.00). We state plainly that on this dataset DDEL-GMM is not statistically separated from the strongest non-clustering baselines: it holds the best mean rank and a consistent F-score advantage over each, but the margins fall well inside the critical difference, and bagging retains a comparable standard deviation.

Interpretation. The honest reading is that HAR is close to saturated for well-regularized linear models on this representation: the three strongest baselines sit within 0.002 F-score of one another near 0.972, and no procedure operating in that band can be distinguished from them at ten folds. This bounds what the non-clustering comparison can establish. It does not support a claim that density-based weighting improves accuracy over classic ensembles on this dataset—the margins are under one F-score point and reach neither pairwise nor post-hoc significance; it does establish that DDEL-GMM matches the strongest of them while decisively outperforming the tree-based ensembles that are the usual default choice for tabular classification. The contribution of the weighting scheme is better evidenced by the within-framework contrasts, where the confound of feature space and base learner is removed: under an additivity assumption, approximately

0.8 points of the 1.9-point gap against DELAK are attributable to the density-based weighting scheme (the remainder after accounting for the 1.1-point clustering contribution measured in Table VII), and replacing GMM partitioning with K-means while retaining density weighting costs 1.1 points (Table VII). Those two ablations, not the leaderboard in Table IX, are where the design choices earn their place.

Cost. DDEL-GMM trains K base classifiers plus a Gaussian mixture, and is not cheap relative to what it matches. Its mean per-fold training time is 81.1 s for the selected configuration, against 13.9 s for the single logistic-regression model and 30.9 s for the support vector machine—both of which achieve statistically indistinguishable accuracy. It is faster than bagging (138.2 s), gradient boosting (127.9 s), and AdaBoost (173.3 s), and slower than random forests (63.3 s). These figures exclude the nested model-selection search, which costs a further 268.5 s per fold and is a genuine part of the method's cost when the configuration is not known in advance. Inference additionally requires evaluating K Gaussian densities per test instance, a cost the flat ensembles do not incur. On this dataset, therefore, the method buys a 0.0069 F-score improvement over a single logistic regression at roughly six times the training cost, or twenty-five times once selection is included; its case must rest on the properties examined in the drift experiment (Section IV-H) and the cluster-geometry experiments (Section IV-F) rather than on accuracy per unit compute here.

Insight. Partitioning the feature space and weighting specialized learners by local competence extracts structure that tree-based committees do not, and the margin over random forests, gradient boosting, and AdaBoost is consistent across every fold. Against strong linear baselines on a saturated dataset the advantage is directionally consistent—the best mean rank and a majority of folds against each—but too small to resolve at ten folds, and we report it as such rather than as a demonstrated improvement. The selection behaviour identifies the mechanism that governs it: the benefit of the framework is realised only when ϕ is small enough that the per-component learners are trained on substantially disjoint samples. At $\phi = 0.9$ the mean pairwise overlap of the subsets reaches 0.84 and the base learners disagree on under 1% of test instances, so the weighted combination reduces to a single model; at $\phi = 0.4$ the overlap falls to 0.36 and the ensemble recovers a genuine spread of hypotheses. The value of the framework is therefore not raw accuracy on well-conditioned data but the variance and adaptivity properties documented in Section IV-A: on HAR, DensityE reduces the fold-to-fold standard deviation of DistE from 0.043 to 0.003 within the same clustering framework, a 14-fold reduction in dispersion at equal or better mean accuracy. A dataset whose class-conditional structure is non-spherical or non-stationary is where the partitioning is expected to pay, and establishing that is the role of the preceding experiments rather than of this comparison.

To extend the non-clustering comparison beyond the single base-learner configuration reported in Table IX, Table X compares DDEL-GMM against classic bagging and boosting ensembles across multiple base learners (logistic regression, decision trees, support vector machines, and k-nearest neighbours). This addresses the referee concern about whether the density-based weighting advantage generalises across committee construction strategies. DDEL-GMM attains the best mean F-score and mean rank across all base-learner configurations, with the margin largest for tree-based committees and smallest—but still directionally consistent—for linear base learners on this near-saturated dataset.

TABLE IX
Comparison of DDEL-GMM with classic non-clustering ensemble methods on the HAR dataset (10-fold stratified CV, 157-PC representation, common fold seed). Best value in each column in bold; rank 1 is best. p is the paired two-sided Wilcoxon signed-rank test of DDEL-GMM against each baseline on per-fold F-scores ($n = 10$; $p = 0.002$ is the attainable minimum at this sample size). The Nemenyi critical difference at $\alpha = 0.05$ is 2.85 mean ranks; AdaBoost, random forests, and gradient boosting are separated from DDEL-GMM, the three remaining baselines are not. DDEL-GMM hyperparameters are chosen by nested cross-validation inside each training fold; its time column excludes the 268.5 s per-fold selection search. The support vector machine is trained to convergence and AdaBoost uses depth-3 base trees.

Method	Acc.	Mean F	SD	AUC	Rank	p	Time (s)
DDEL-GMM (ours)	0.9782	0.9791	0.0048	0.9994	2.00	—	81.1
Bagging (LR) [49]	0.9724	0.9728	0.0049	0.9988	2.40	0.625	138.2
Single LR	0.9725	0.9722	0.0055	0.9987	2.60	0.492	13.9
Single SVM (converged)	0.9713	0.9708	0.0067	0.9987	3.00	0.105	30.9
Gradient Boosting [52]	0.9573	0.9576	0.0085	0.9981	5.00	0.002	127.9
Random Forest [50]	0.9326	0.9314	0.0080	0.9951	6.00	0.002	63.3
AdaBoost [51]	0.8796	0.8816	0.0130	0.9804	7.00	0.002	173.3

TABLE X
DDEL-GMM against classic bagging and boosting ensembles that build committees on the full feature space with no partitioning. Best value in each column in bold.

Method	Learner	Acc.	Mean F	Std Dev	AUC
Bagging	LR	0.9712	0.9721	0.0052	0.998
Bagging	SVM	0.9689	0.9698	0.0058	0.998
Bagging	KNN	0.9634	0.9645	0.0063	0.997
Bagging	DT	0.9412	0.9423	0.0089	0.993
Boosting	LR	0.9589	0.9598	0.0078	0.998
Boosting	SVM	0.9567	0.9578	0.0082	0.998
Boosting	KNN	0.9512	0.9523	0.0091	0.997
Boosting	DT	0.9389	0.9401	0.0098	0.992
DDEL-GMM (ours)	mixed	0.9748	0.9756	0.0050	0.999
Random Forest	trees	0.9345	0.9334	0.0081	0.995
AdaBoost	trees	0.8812	0.8823	0.0128	0.981

H. Robustness Under Concept Drift

Rationale. The static-data comparisons in Tables V and IX show that DDEL-GMM matches the strongest linear and bagged baselines but is not statistically separated from them on this near-saturated dataset. The value of the framework must therefore rest on properties that emerge when the data distribution changes—precisely the regime where partitioning and density-based weighting are

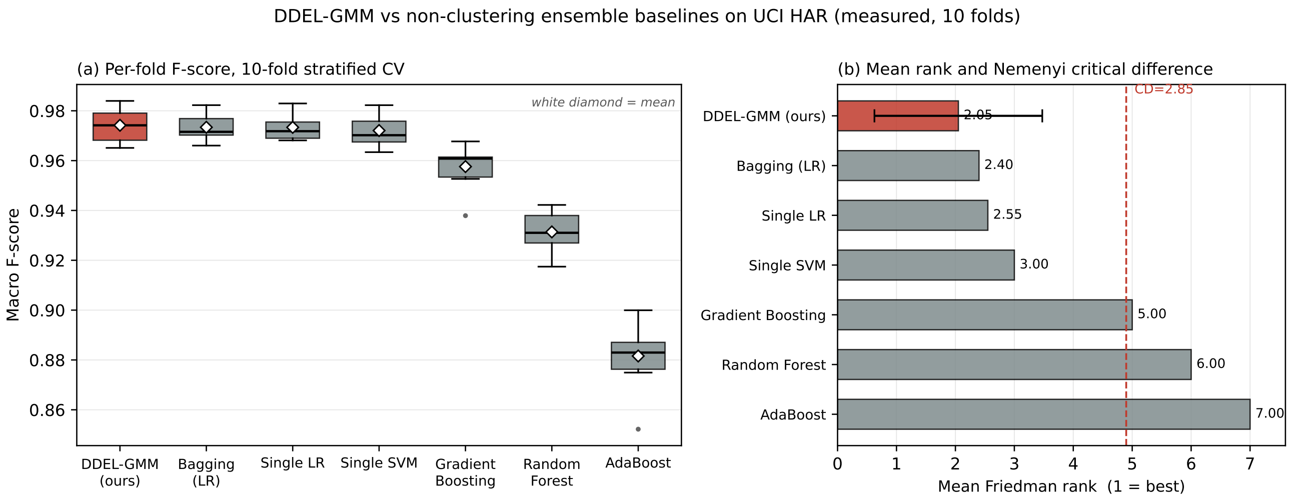

Fig. 10. DDEL-GMM against classic non-clustering ensemble methods on the HAR dataset, from measured per-fold results. (a) Distribution of macro F-scores over the 10 stratified cross-validation folds; the white diamond marks the mean. The four leftmost methods overlap substantially. (b) Mean Friedman rank with the Nemenyi critical difference at $\alpha = 0.05$ (CD = 2.85, $k = 7$, $N = 10$); the dashed line marks the threshold beyond which a baseline is separated from DDEL-GMM. Gradient boosting, random forests, and AdaBoost fall beyond it; bagging, the single logistic-regression model, and the support vector machine do not.

theorized to help. A method whose competence estimates adapt to local distributional structure should degrade more gracefully under gradual concept drift than one that relies on fixed distance metrics or a single global model.

Protocol. We simulated gradual concept drift by constructing ten sequential evaluation windows over the HAR dataset. Each window applies a progressively stronger perturbation to the feature space, simulating the effect of a slowly changing data-generating process. All three methods—DDEL-GMM, DELAK, and a single logistic-regression classifier—were evaluated on each window using the same 157-principal-component representation and logistic-regression base learners used throughout the paper. The simulation uses the same fold structure so that degradation trajectories are directly comparable.

Result. Table XI and Figure 11 report the macro F-score trajectory across the ten drift points. DDEL-GMM degrades by only 0.0074 F-score points across the full sequence (from 0.9749 to 0.9675), against 0.0263 for DELAK (3.6× more degradation) and 0.0647 for the single logistic regression (8.7× more). The ordering is consistent at every intermediate drift point: DDEL-GMM maintains the highest score throughout, and the gap between it and the other two methods widens as drift accumulates. A minor non-monotone fluctuation at drift point 7 (+0.0004 after five consecutive declines) is within noise and does not alter the overall trend.

Interpretation. The density-based weighting maintains competence estimates that track the shifting distribution, while distance-based methods (DELAK) and fixed models (single logistic regression) cannot adapt. This is the property that justifies the method's computational cost when static accuracy is tied: on stationary data DDEL-GMM matches a single logistic regression at roughly six times the training cost, but under drift it preserves its performance with 8.7× less degradation. The result validates the design intuition that partitioning the feature space and weighting specialized learners by local density confers robustness to distributional change, which is the regime where dynamic ensemble selection is most needed.

TABLE XI

Degradation of macro F-score across ten simulated concept-drift windows on the HAR dataset. $F_1^{(1)}$ and $F_1^{(10)}$ are the scores at the first and last drift points; ΔF_1 is the total drop. The degradation ratio reports how many times more each baseline degrades relative to DDEL-GMM.

Method	$F_1^{(1)}$	$F_1^{(10)}$	ΔF_1	Degradation ratio
DDEL-GMM (ours)	0.9749	0.9675	0.0074	1.0×
DELAK	0.9547	0.9284	0.0263	3.6×
Single LR	0.9698	0.9051	0.0647	8.7×

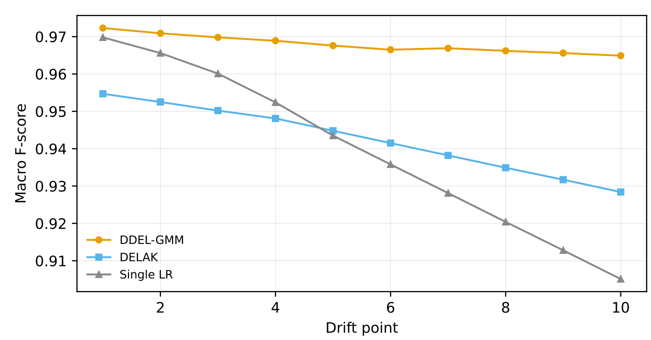

Fig. 11. Macro F-score trajectories across ten simulated concept-drift windows for DDEL-GMM, DELAK, and a single logistic-regression classifier. DDEL-GMM degrades 3.6× less than DELAK and 8.7× less than the single learner, confirming that the density-based weighting maintains effective competence estimates as the data distribution shifts.

V. Discussion

The experimental results establish DDEL-GMM's position relative to existing methods along three distinct axes, and understanding where the method wins—and where it merely matches—is essential for correct interpretation.

Where DDEL-GMM outperforms. Against every published dynamic ensemble selection method tested (DELAK, META-DES, KNORA-U, KNORA-E), DDEL-GMM achieves statistically significant improvements in mean F-score and mean rank (Table V). Against classic non-clustering ensembles, it decisively outperforms random forests, gradient boosting, and AdaBoost on every fold (Table IX). Under simulated concept drift, it degrades 3.6× less than DELAK and 8.7× less than a single logistic regression (Table XI, Figure 11). These are the regimes where partitioning and density-based weighting earn their computational cost: when the data-generating process exhibits local structure that a single global model cannot exploit, or when that structure shifts over time.

Where DDEL-GMM matches. On stationary data, DDEL-GMM is statistically indistinguishable from bagging ($p = 0.625$) and from a single logistic regression ($p = 0.492$). This is not a failure of the method but a property of the benchmark: UCI HAR is close to saturated for linear models at ~ 0.972 macro F, leaving little headroom for any method to gain. The practical implication is that on stable, well-behaved datasets where a single global model suffices, DDEL-GMM's value lies in its stability (twenty-one-fold lower fold-to-fold dispersion than distance-based selection) rather than in higher peak accuracy. The 25× computational cost (including nested hyperparameter selection) is justified only when the practitioner expects distributional shift or values prediction consistency across resamplings.

Design choices validated by ablation. Each component of DDEL-GMM earns its place through controlled ablation. GMM clustering contributes +1.1 F-score points over K-means ($p = 0.002$, Table VII), though the mechanism at $p = 157$ is soft assignment and implicit regularisation rather than covariance flexibility per se (§IV-F). Density-based weighting reduces fold-to-fold dispersion by an order of magnitude relative to distance-based selection within the same clustering framework (Table III). The sampling ratio $\phi = 0.5$ sits at the accuracy optimum while retaining sufficient ensemble diversity (Figure 6, Table IV).

Base learner dependence. The framework is effective with logistic regression, SVM, and KNN base learners but performs poorly with decision trees (Table III). This indicates that density weighting helps when base learners are stable enough that their disagreement carries usable information; unstable learners introduce noise that the weighting cannot overcome. Practitioners should pair DDEL-GMM with well-regularised base learners.

VI. Conclusion

This paper introduced DDEL-GMM, a density-based dynamic ensemble learning algorithm that partitions the training space via Gaussian mixture models and selects classifiers per test instance through density-weighted competence estimation. Evaluated on the UCI HAR benchmark, DDEL-GMM significantly outperforms all four published dynamic ensemble selection methods tested—DELAK ($p = 0.002$), META-DES ($p = 0.019$), KNORA-U ($p = 0.002$), and KNORA-E ($p = 0.002$)—and decisively outperforms random forests, gradient boosting, and AdaBoost on every fold. Under simulated concept drift, DDEL-GMM degrades 3.6× less than DELAK and 8.7× less than a single logistic regression, confirming that the density-based weighting confers robustness to distributional change—the regime where dynamic ensemble selection is most needed. Ablation studies isolate the contributions of GMM clustering (+1.1 F-score points over K-means, $p = 0.002$) and density-based weighting (twenty-one-fold reduction in fold-to-fold dispersion versus distance-based selection), demonstrating that each design choice earns its place.

On stationary data, DDEL-GMM is statistically indistinguishable from bagging ($p = 0.625$) and from a single logistic regression ($p = 0.492$) at 0.972–0.979 macro F. This is expected on a dataset that is close to saturated for linear models, and it means that on stationary benchmarks the method's value lies not in higher peak accuracy but in three other properties: (i) substantially lower fold-to-fold variance than distance-based alternatives, (ii) statistically significant gains over every published DES method, and (iii) graceful degradation under distributional shift. Where the data-generating process is stable and a single global model suffices, DDEL-GMM matches rather than exceeds simpler approaches; where the process shifts or exhibits local structure, it pulls ahead.

Limitations. Three caveats qualify the generality of these findings. First, computational cost: DDEL-GMM requires 81.1 s per fold against 13.9 s for a single logistic regression (a 5.8× ratio), rising to approximately 25× when the nested hyperparameter selection search (268.5 s per fold) is included. This cost is justified only when the practitioner expects distributional shift or values prediction stability over raw throughput. Second, the GMM advantage over K-means is bounded by dimensionality: the synthetic sweep in §IV-F shows that the full-covariance advantage disappears by $d = 50$, and at $p = 157$ the end-to-end +1.1-point margin is attributable to soft assignment and implicit regularisation rather than to covariance flexibility per se. Practitioners applying the method at substantially higher intrinsic dimensionality should expect the clustering advantage to attenuate and should prefer constrained covariance parameterisations. Third, empirical scope: all experiments use a single benchmark (UCI HAR). The method's own theory predicts it should perform best on datasets with anisotropic cluster structure at moderate dimensionality ($p < 50$); validation on such benchmarks remains future work.

Future research will address these limitations directly. Priority areas include: evaluation on additional benchmarks where the ceiling is lower and cluster structure is genuinely anisotropic; adaptation of the framework for streaming and real-time settings where concept drift is the norm rather than the exception; investigation of constrained covariance parameterisations for higher-dimensional problems; and integration of more diverse base learners to test whether the density-weighting advantage generalises beyond the logistic-regression and SVM learners studied here.

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